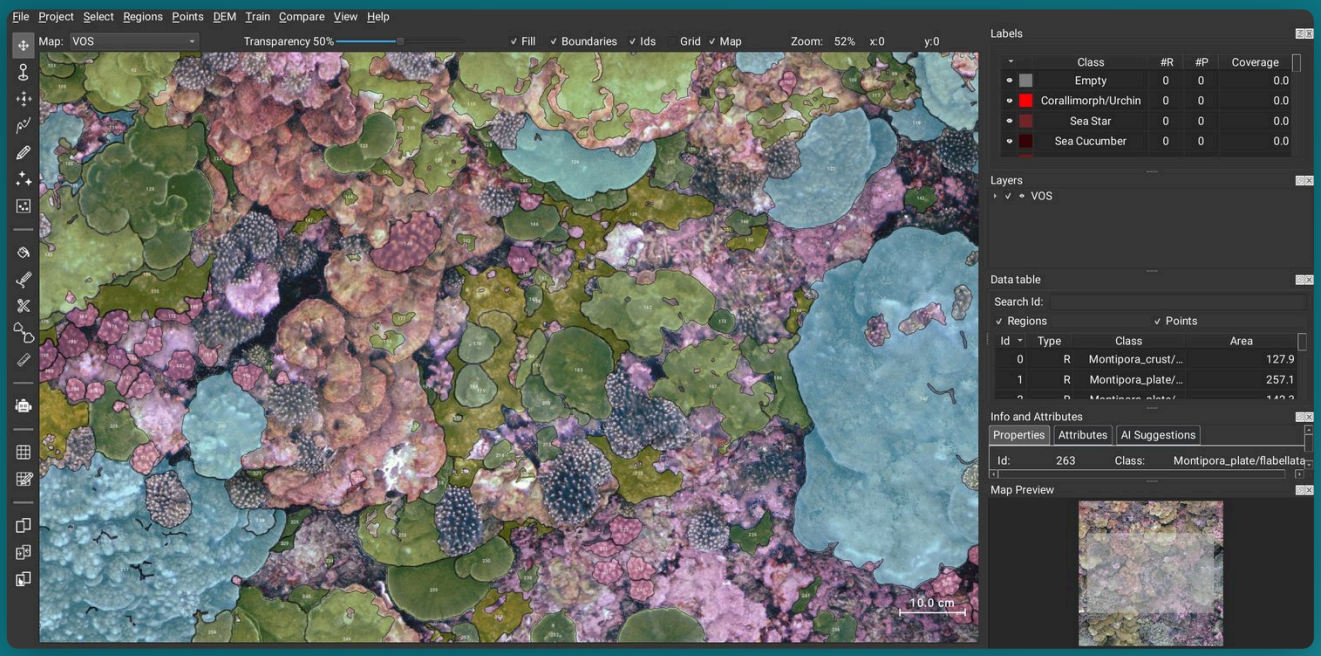
FIU CORAL REEF FISHERIES LAB · STANDARD OPERATING PROCEDURE

TagLab

Annotating reef orthomosaics: a step-by-step guide for new lab members



Software: TagLab, FIU lab version (based on TagLab 2026.4.1, CNR-ISTI Visual Computing Lab)
Covers: projects and maps · annotation tools · points · exports · DEM metrics · year-to-year comparison · AI training and classification · CoralNet
Version 1.0 · November 2026 · Companion document: *TagLab Quick Reference*

About this guide

Purpose

This guide explains how the lab uses TagLab to outline and classify colonies on reef orthomosaics, measure them (area, surface area, DEM metrics), follow them from one survey to the next, train and apply AI classifiers, and prepare CoralNet training data. It is written for new lab members: read chapters 1 to 4 before your first annotation session, and the other chapters when you need them. People who already annotate can use the *TagLab Quick Reference* instead.

Scope

- **Software:** the lab's version of TagLab, from the GitHub repository `nirivas/TagLab-FIU`. It is the official TagLab 2026.4.1 plus the lab's changes (listed in `CHANGES.md` in the TagLab folder). Menus and buttons in this guide are those of that version; the official TagLab and its online manual differ in places.
- **Data:** the lab's plots (Culebra year-to-year plots, PEV plots with an orthomosaic and a DEM, the sponge nursery).
- **Not covered:** making the orthomosaics and DEMs (Metashape, Viscore), and the statistics done afterwards in R.

Responsibilities

Who	Does
Annotator	Annotates maps following this guide, saves often, keeps the project and its images together, reports problems with `TagLab.log`.
Project lead for a plot	Sets up the project (maps, pixel size, dictionary, working area), checks the annotations, runs the year-to-year comparison and the exports.
Lab member in charge of TagLab	Keeps TagLab up to date on the lab computers, trains the classifiers on the GPU computer, updates this guide.

Conventions

- Menu commands are written like this: **File ▸ Save Project**.
- Keys are written like this: Ctrl+S. **SHIFT + click** means: hold SHIFT and click with the left mouse button (LMB). **SHIFT + right click** uses the right button (RMB).
- Toolbar buttons are shown with their icon, for example  *Positive/negative clicks segmentation*.
- The keys given are the defaults. Every key can be changed (**File ▸ Keyboard and Mouse Shortcuts...**); if a lab computer has different keys, the menus and the Help window show them.

> **About the screenshots**
> The screenshots were taken from TagLab itself, on the sample projects that come with TagLab (CC BY-NC-SA 4.0: orthoimages acquired and annotated by the Stuart Sandin Lab, Scripps Institution of Oceanography; the DEM sample courtesy of the Moorea Island Digital Ecosystem Avatar project). Their labels are Pacific corals (*Pocillopora*, *Montipora*, *Porites*). The lab's own dictionaries (FDEP, sponge nursery) work in exactly the same way.

Revision history

Version	Date	Changes
1.0	November 2026	First version, for the FIU lab version of TagLab (based on 2026.4.1, with the lab changes up to November 2026).

Contents

About this guide 2

1. Before you start 6
- 1.1 What TagLab does 6
- 1.2 The lab computers 6
- 1.3 Installing, starting and updating 6
- 1.4 The files TagLab uses 7
- 1.5 Rules for safe work 7

2. The TagLab window 8
- 2.1 A tour of the window 8
- 2.2 Moving around the map 8
- 2.3 How the tools are used: SHIFT, SPACE and ESC 9
- 2.4 The toolbar 9
- 2.5 Help inside TagLab 10

3. Setting up a project 12
- 3.1 Create the project and add a map 12
- 3.2 Checking the pixel size with the Ruler 12
- 3.3 The Maps Editor 13
- 3.4 Labels and dictionaries 13
- 3.5 Region attributes 15
- 3.6 The working area 15
- 3.7 Saving and autosave 16
- 3.8 Moving a project to another computer 16

4. Annotating regions 17
- 4.1 Which tool when 17
- 4.2 Selecting regions 17
- 4.3 Positive/negative clicks (RITM) 17
- 4.4 4 clicks (DEXTR) 18
- 4.5 Freehand 18
- 4.6 SAM (Segment Anything) 19
- 4.7 Edit Border 20
- 4.8 Cut 20
- 4.9 Assigning classes 20
- 4.10 Move Region 21
- 4.11 Region operations (Regions menu and right click) 22
- 4.12 Undo and redo 23
- 4.13 Class, note and attributes of a region 23
- 4.14 Other tools 23
- 4.15 SOP: annotating a plot 24

5. Points 25
- 5.1 Placing points by hand 25
- 5.2 Sampling points on a map 25
- 5.3 Labelling points from the regions 26
- 5.4 Importing points 27

6. Checking and exporting 28
- 6.1 The Data table 28
- 6.2 Selecting by properties 28
- 6.3 Shape measurements 29

6.4 Exports 29

7. DEM: surface area and 3D metrics 31

7.1 Seeing the DEM 31
7.2 Surface area 31
7.3 Colony DEM metrics 31
7.4 Plot metrics (plot rugosity) 32
7.5 SOP: DEM metrics of a plot 33

8. Year-to-year comparison 34

8.1 The workflow 34
8.2 Aligning a new survey: the Alignment Tool 34
8.3 Split screen 37
8.4 Automatic matches 38
8.5 Carrying regions over to a new survey 39
8.6 The comparison table 42
8.7 Exporting the year-to-year data 44
8.8 SOP: annotating a new year 44

9. AI: training a network and classifying maps 45

9.1 Exporting a training dataset 45
9.2 Merging classes on export 46
9.3 Training a network (GPU computer) 47
9.4 Reading the training results 48
9.5 Classifying a map 48
9.6 Good practice for training data 49

10. CoralNet 50

10.1 Training a CoralNet source with your regions 50
10.2 The CoralNet Toolbox 50

11. Settings and shortcuts 52

11.1 Settings 52
11.2 Keyboard and mouse shortcuts 52

12. Troubleshooting 53

12.1 When something goes wrong 53
12.2 Common problems 53
12.3 Keeping TagLab up to date 53

Appendix A. Default keys 54

Appendix B. Every menu command 55

Appendix C. Units and formulas 56

Appendix D. Columns of the data table export 57

Glossary 58

1. Before you start

1.1 What TagLab does

TagLab is a program for annotating large images of the reef. You open an **orthomosaic** of a plot (one top-down image stitched from hundreds of photos), outline each colony as a **region**, give it a **class** (its species or category), and TagLab measures it. With the **DEM** of the same survey it also measures surface area, height, volume and rugosity. With two surveys of the same plot it follows each colony through time (growth, shrinkage, death, recruitment, fission and fusion).

Most outlines are not drawn by hand: AI tools draw them from a few clicks (RITM, the *4 clicks* tool, SAM), and a trained network can classify a whole map automatically. You then check and correct the result.

1.2 The lab computers

	Annotation computers	GPU computer
Processor for AI	CPU only	NVIDIA GPU (CUDA)
Use it for	Annotating (RITM, 4 clicks, freehand, editing), points, exports, DEM metrics, the Alignment Tool, Carry Regions Over, exporting training datasets	Training networks (Train Your Network), classifying whole maps, SAM
Speed	RITM: about 0.5 s per click. Classifying a 1500 × 1500 px area: about 2 minutes. SAM: very slow.	Training and classification many times faster

Both run the same TagLab version and open the same project files.

1.3 Installing, starting and updating

TagLab is already installed on the lab computers. To install it on a new computer, follow `SETUP_NEW_COMPUTER.md` in the TagLab folder. In short:

1. Install **Python 3.11 (64-bit)** from python.org, with *Add python.exe to PATH* ticked.
2. On https://github.com/nirivas/TagLab-FIU click **Code > Download ZIP** and extract it, for example to `C:\TagLab-FIU`.
3. Double-click **`install_windows.bat`**. It creates TagLab's own Python environment (the `venv` folder), installs the packages (the GPU version of PyTorch when the computer has an NVIDIA card) and downloads the AI models (about 1.4 GB). At the end it asks whether to download SAM (2.5 GB; only useful with an NVIDIA GPU).
4. Start TagLab with **`opentaglab.bat`** (right-click it > Send to > Desktop to make a shortcut).

Updating. When a newer lab version is on GitHub, TagLab shows a notice at start. Close TagLab and double-click **`update.bat`**. It replaces only the files that changed (the old ones are kept in `update_backups\`) and never touches your projects, your data, the models or the networks you trained. If it says that the required packages changed, run `install_windows.bat` again.

> **Use the lab version**
> The official TagLab (from the CNR-ISTI website) does not have the lab's fixes and tools, and its autosave can close the program. Open lab projects with the lab version.

1.4 The files TagLab uses

File	What it is	Where
Project (`.json`)	Your work: the maps, every region and point, the labels, the comparison tables	Wherever you save it; keep it next to the images
Orthomosaic (`.tif` , `.png` , `.jpg`)	The image of a survey	Your data folder; the project stores its path
DEM (`.tif`)	The heights of a survey	Next to the orthomosaic
`<project>_autosave.json`	Autosave copy	Next to the project
Dictionary (`FDEP.json` , `spongenursery.json`)	A list of labels and colours	TagLab folder
`TagLab.log`	The log: errors and results of long operations	TagLab folder (rewritten at every start)
`config.json`	TagLab's settings and the list of classifiers	TagLab folder
`models\`	The AI models and the trained classifiers (`.net`)	TagLab folder

The project file contains the **paths** of the images, not the images. If you move or rename the image files, TagLab asks you to locate the first missing one when you open the project; the others in the same folder are then found automatically. Save the project afterwards.

1.5 Rules for safe work

1. **Never edit the original orthomosaics or DEMs.** TagLab never changes them: the Alignment Tool writes new files (suffix `_coreg`) and leaves the originals as they are.
2. **One project per plot.** All the surveys of a plot go into the same project, so they can be compared.
3. **Save often** with Ctrl+S, and **turn on autosave** (File ▸ Settings.., see section 3.7).
4. **Don't open the same project on two computers at the same time.** The last save wins and the other person's work is lost.
5. **OneDrive / SharePoint folders:** TagLab saves safely even while OneDrive syncs (the old file stays intact if a save fails), but wait for the sync to finish before opening the project on another computer.
6. **Before big operations** (Carry Regions Over, classification, import of a label map), save the project. They can be undone with Ctrl+Z, but a saved copy is the safest undo.
7. **When something goes wrong**, don't close TagLab at once: read the message, then copy `TagLab.log` (it is rewritten at the next start) and send it with a description of what you did.

2. The TagLab window

2.1 A tour of the window

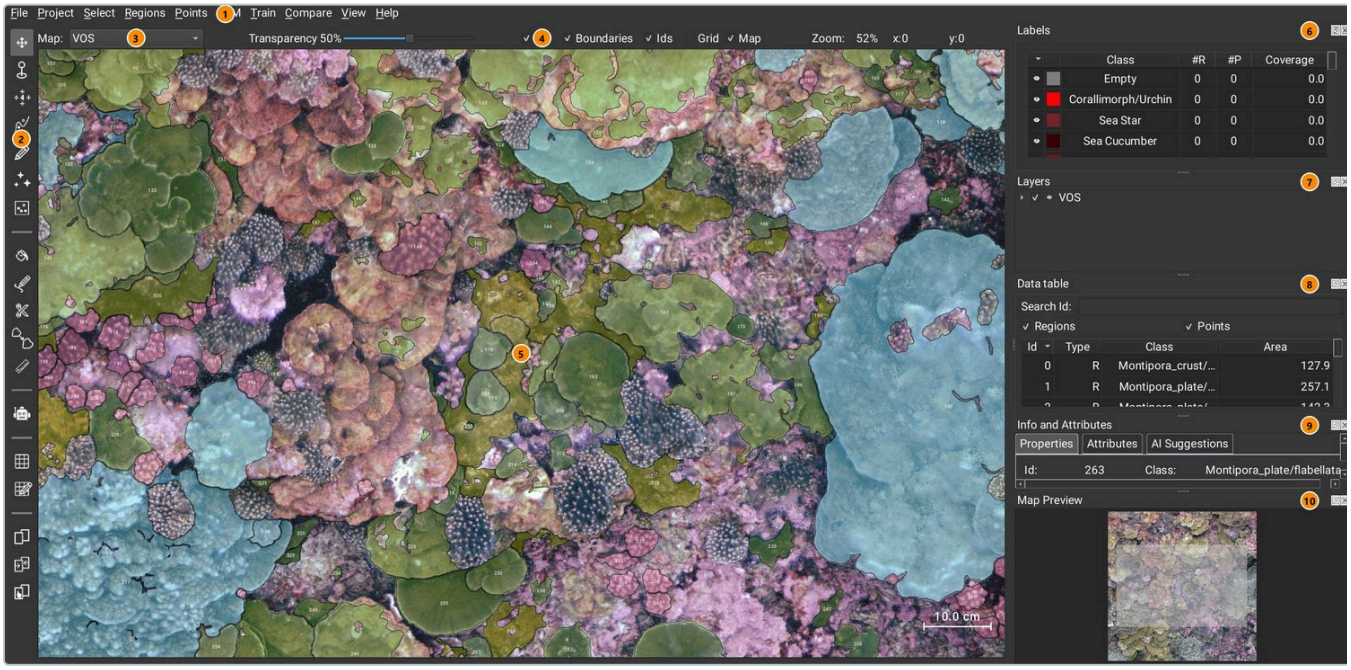


Figure 1. The TagLab window with a map open. The numbers are explained below.

1. **Menu bar:** File, Project, Select, Regions, Points, DEM, Train, Compare, View, Help.
2. **Toolbar:** the annotation tools (section 2.4). The button of the current tool is highlighted.
3. **Map:** the map shown (choose another survey of the project here). In split screen there are two of these lists, one per map.
4. **Header bar:** what is drawn over the map (Fill, Boundaries, Ids, Grid, Map), the transparency of the fill, the zoom and the position of the mouse in map pixels.
5. **The map viewer.** The panel of instructions of the current tool appears at its bottom left.
6. **Labels:** the classes of the project, with the number of regions (#R) and points (#P) and the area covered.
7. **Layers:** the maps and the extra layers (grid, shapes, sampling areas) that can be shown or hidden.
8. **Data table:** the list of the regions and points of the map. In split screen the **comparison table** takes its place.
9. **Info and Attributes:** the properties of the selected region (id, class, area, perimeter, surface area, note) and its attributes.
10. **Map Preview:** the whole map, with the part shown in the viewer; click or drag in it to move there.

The panels (6–10) can be moved, floated or closed; **View** brings them back.

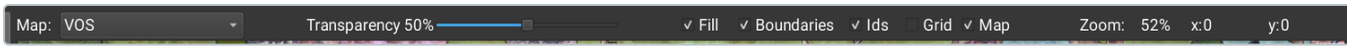


Figure 2. The header bar. Transparency changes how much the fill of the regions hides the image.

2.2 Moving around the map

Action	How
Zoom in / out	Mouse wheel, centred on the mouse
Pan (move the view)	Drag with the left button (Pan/Zoom tool), or Ctrl + drag with any tool

Action	How
Jump to a place	Click or drag in **Map Preview**
Go to a region by id	Type the id in *Search Id* of the Data table, then click its row
Show / hide the fill, borders, ids, grid	Q, W, E, G or the header bar checkboxes
See the DEM instead of the image	C (DEM ▸ Switch RGB/DEM), when the map has a DEM

> **Ids**
> Turn the ids on (E) when you check a table or a CSV against the map: the number drawn on each region is its id, the same as in the Data table, the comparison table and the exports.

2.3 How the tools are used: SHIFT, SPACE and ESC

All the drawing tools follow the same rules:

- **The mouse acts on the map only with SHIFT held.** A plain drag always pans the view, so you can move around without leaving the tool. SHIFT + click adds a point, SHIFT + drag draws, SHIFT + right click adds a negative point.
- Space **confirms** what the tool has prepared (a new outline, a cut, a border edit).
- Esc **cancels** the operation in progress.
- Ctrl+Z removes the last point while a tool is working, and undoes the last operation otherwise (Ctrl+Shift+Z redoes).
- **The instructions panel** at the bottom left of the map always lists the clicks and keys of the current tool.
- **Double click** on a region selects it in the tools that work on one region (Edit Border, Cut).

2.4 The toolbar

The toolbar buttons, from top to bottom. Keys can be changed with File ▸ Keyboard and Mouse Shortcuts....

	Button	Key	What it does
	Pan/Zoom	1	Pan and zoom the map; click a region to select it (SHIFT: add to the selection), SHIFT + drag: select with a rectangle.
	Place annotation point		Place an annotation point (SHIFT + click).
	4-clicks segmentation	2	Outline a colony from 4 clicks on its top, bottom, left and right edge (DEXTR).
	Positive/negative clicks segmentation	3	Outline a colony with positive (inside) and negative (outside) clicks (RITM). Also corrects a selected region.
	Freehand segmentation	4	Draw the outline by hand, then SPACE.
	Watershed segmentation (hidden here)		Scribbles inside and outside a colony (shown only in the Digital Heritage research field).
	SAM - all instances in an area		SAM: outline every object in a square work area (slow without an NVIDIA GPU).
	SAM - positive/negative clicks in an area		SAM with positive / negative clicks in a work area.
	Rows analysis (hidden here)		Rows analysis (Digital Heritage only).

	Button	Key	What it does
	Assign class to region	5	Give the class chosen in the Labels panel to a region (SHIFT + click).
	Edit region border	6	Redraw part of the border of the selected region.
	Cut region	7	Cut the selected region in two along a line.
	Move region (SHIFT + drag or the arrow keys; SPACE re-fits the outline)		Move, turn and correct regions (drag, arrows, , and ., points).
	Measure tool	9	Measure a distance; set the pixel size from a known length.
	Fully auto semantic segmentation	0	Classify the map with a trained network (Select Classifier window).
	Create grid		Create a grid of cells on the map (to annotate cell by cell).
	Active/disactive grid operations		Turn grid operations on or off (cell states and notes).
	Split screen	Alt + S	Show two maps side by side (year-to-year).
	Compute automatic matches		Match the colonies of the two maps automatically.
	Edit matches		Edit the matches by hand.

Watershed and Rows appear only when the research field in **File ▸ Settings..** is *Digital Heritage*; the lab uses *Marine Ecology*.

2.5 Help inside TagLab

- **Help ▸ Help** (Ctrl + H) lists the main keys (it shows the keys set on that computer).
- The instructions panel describes the current tool.
- Hold the mouse over a button to see its name.
- `CHANGES.md` in the TagLab folder lists everything the lab version changes.

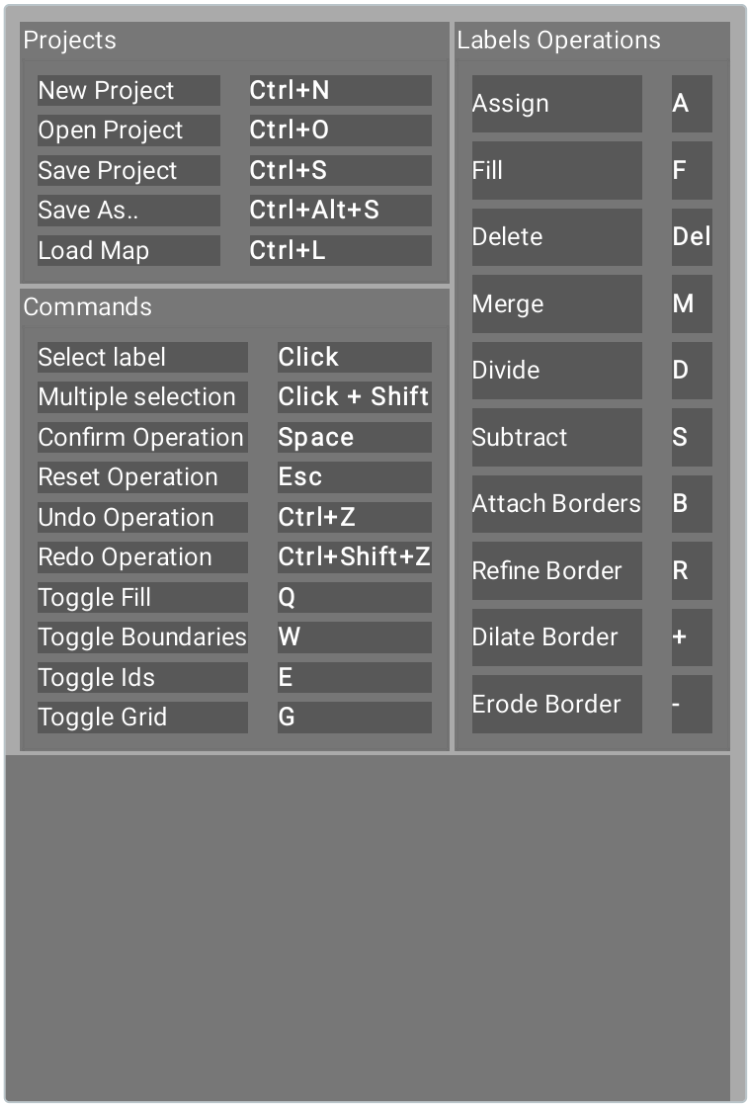


Figure 3. The Help window (Ctrl+H) with the main keys.

3. Setting up a project

A project holds all the surveys of one plot. The project lead usually sets it up once; annotators then open it with **File ▸ Open Project** (Ctrl + O).

3.1 Create the project and add a map

1. **File ▸ New Project** (Ctrl + N). The new project gets the labels of the default dictionary (section 3.4).
2. **Project ▸ Add New Map...** (Ctrl + L) and fill in the window (Figure below):
   - **Map Name:** short and unique, with the plot and the year, for example `Plot12_2026`. It appears in the exports and in the colony names (`Plot12_2026:45`).
   - **RGB Image:** the orthomosaic (`...` to browse).
   - **Depth Image:** the DEM of the same survey, if there is one. It must cover the same area as the orthomosaic, with the same number of pixels.
   - **Acquisition Date:** the survey date, `YYYY-MM-DD`.
   - **Pixel size (mm):** the size of one pixel on the reef (see 3.2).
3. **Apply.** Large maps take a while to load.
4. Save the project (**File ▸ Save As**) next to the images, with a clear name (`Plot12.json`).
5. Add the other surveys of the plot the same way (one map per survey).



Figure 4. Add New Map (Map Settings). The same window opens from the Maps Editor to change a map.

> **The pixel size matters**
>
> Every area, perimeter, length and DEM metric in cm depends on the pixel size. TagLab does **not** read it from the GeoTIFF: take it from the photogrammetry report (ground resolution of the orthomosaic) or measure it with the Ruler. Check it in the Maps Editor before you export anything.

3.2 Checking the pixel size with the Ruler

1. Choose the Ruler (9).
2. Zoom on an object of known length (a scale bar, a quadrat side, a tape).
3. SHIFT + click on one end, then SHIFT + click on the other. The *Measure And Set Scale* window shows the length in mm and in pixels.
4. To correct the pixel size, type the **true** length in the mm field: the pixel size is recomputed. Click **Set new pixel size**.

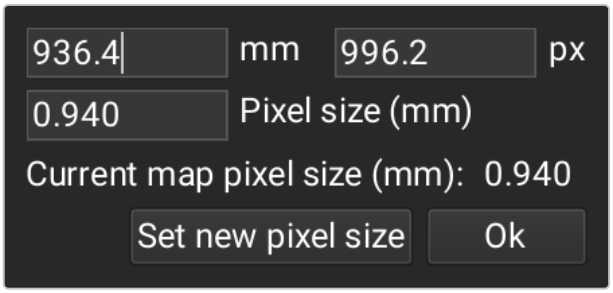


Figure 5. The Ruler window: measured length (mm and px), the pixel size it implies, and the map's current pixel size.

3.3 The Maps Editor

Project ▸ Maps Editor... (Ctrl + M) lists the maps of the project with their date, size, pixel size, whether they are georeferenced and have a DEM, and their number of annotations. **Edit** opens the map settings (name, files, date, pixel size); **Delete** removes the map and its annotations from the project (the image files stay on disk).

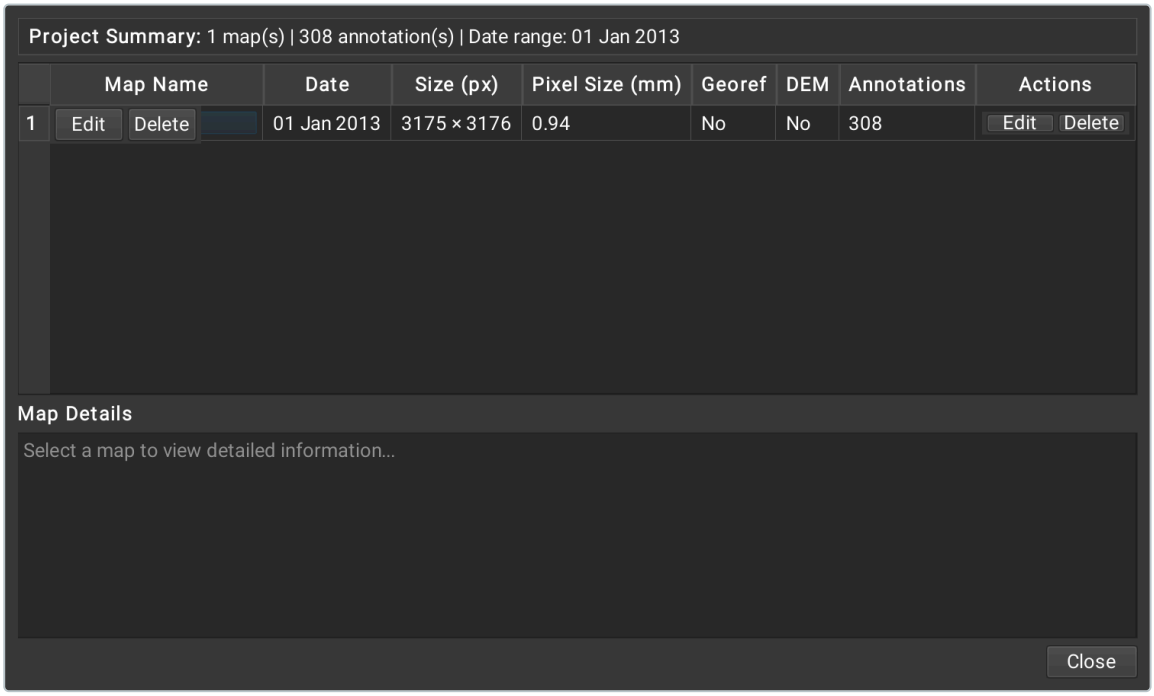


Figure 6. The Maps Editor.

3.4 Labels and dictionaries

The **labels** are the classes of the project. They come from a **dictionary** file, which is copied into the project:

- `FDEP.json`: the FDEP species list (the lab's default for coral plots).
- `spongenursery.json`: the sponge nursery labels (six sponges, each "(Boat)" and "(Scuba)", plus Sand, Macroalgae, Brick and Soft Coral).

To use a dictionary in a project: Project ▸ Labels Dictionary Editor... > **Load**, choose the file, and answer **Replace** (or **Append** to add its labels to the current ones), then **Ok**. In the same window you can add a label (name and RGB colour, **Add**), change one (**Update**) or remove one (**Delete**), and **Save** the list as a new dictionary file. Changing a dictionary file later does not change projects that already use it.

The dictionary used by new projects is set in File ▸ Settings.. (*Default dictionary*).

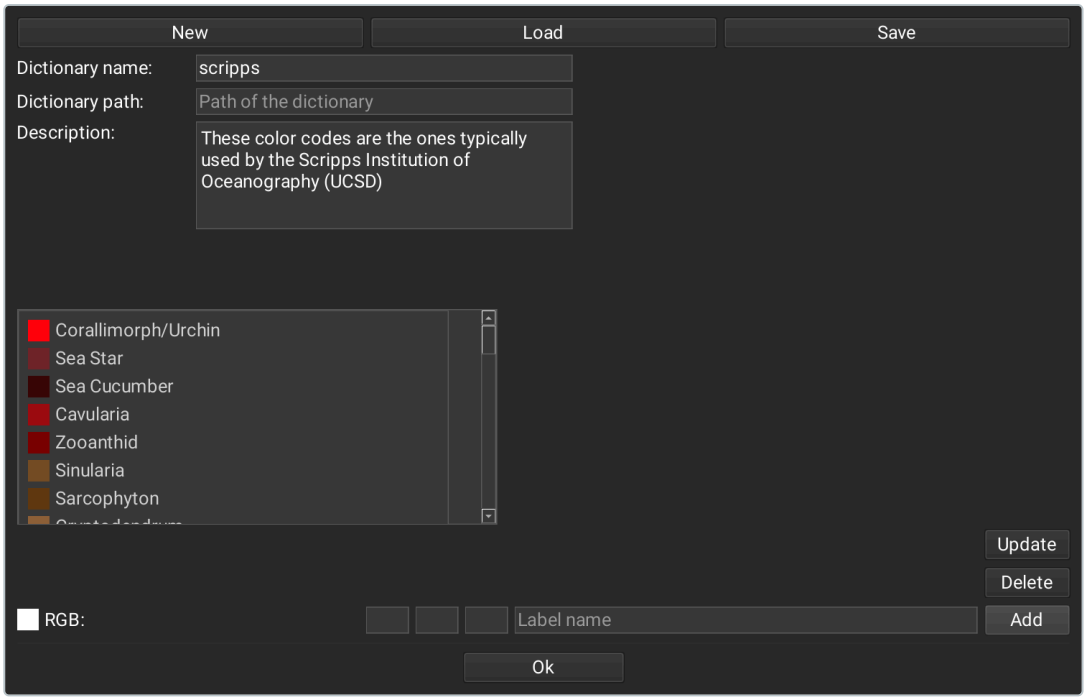


Figure 7. The Labels Dictionary Editor.

> **Colours**
> Every label must have its own colour. Avoid pure black: it is the colour of *Background* in training datasets (FDEP's "Sand" is black; TagLab warns about it when you export a dataset).

The Labels panel

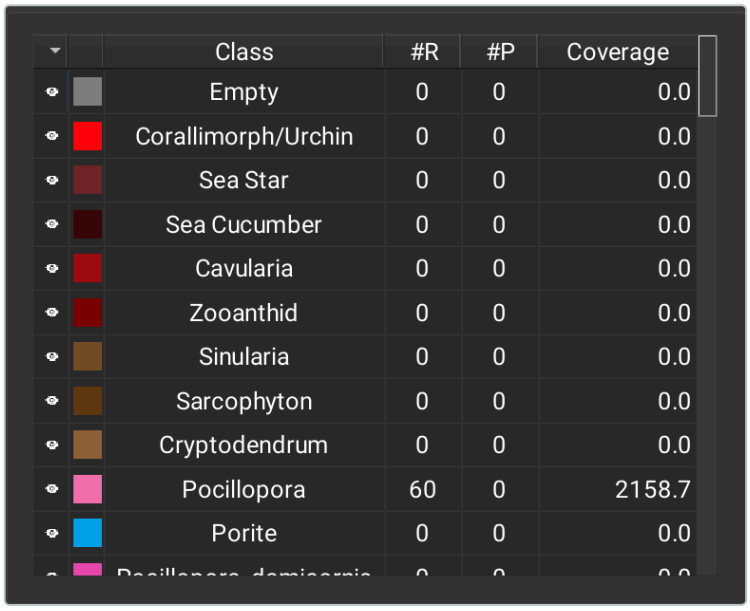


Figure 8. The Labels panel: visibility (eye), colour, class, number of regions (#R) and points (#P), and the area covered (cm^2).

- **Click a class** to make it the active class (used by the Assign tool).
- **Double-click a class** to give it to all the selected regions.
- **The eye** hides or shows the regions of that class.

- Click a column header to sort the classes (for example by number of regions); click the colour header to go back to the dictionary order.

3.5 Region attributes

Region attributes are extra fields every region can have, for example *bleaching* (keywords *none / partial / total*), *disease* (text) or *percent alive* (number). Define them in **Project ▸ Region Attributes...**: type a name, choose a type (string, integer number, decimal number, boolean, keyword), the min and max or the list of keywords, then **Add** and **Apply**. You fill them in for each region in the *Attributes* tab of the Info panel (section 4.13). They are exported as columns of the CSV and of the shapefile.

Some attributes are added by TagLab itself: the DEM metrics (chapter 7) and the carry-over status (chapter 8).

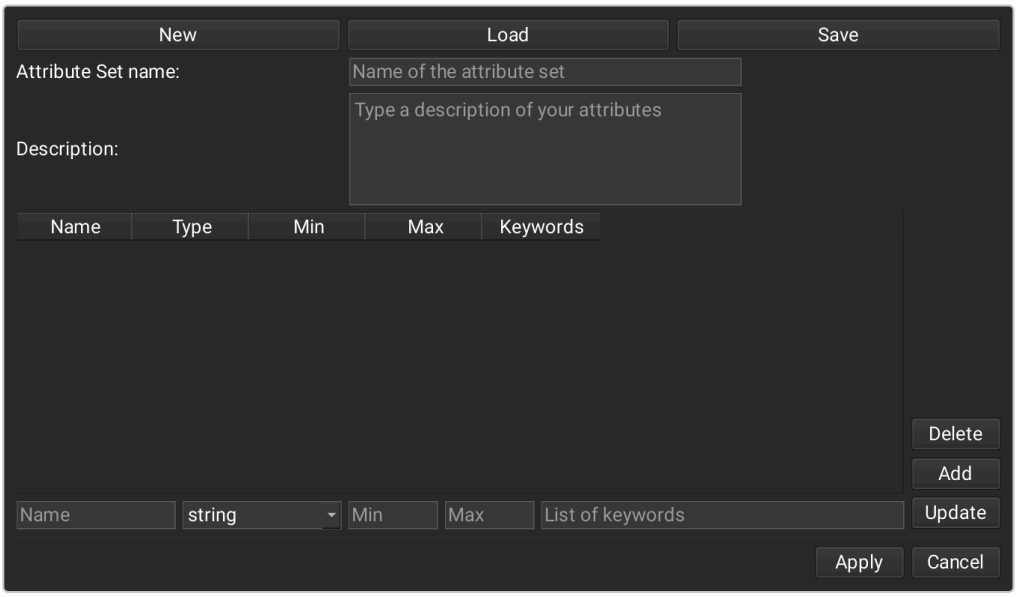


Figure 9. Region Attributes.

3.6 The working area

The working area is a rectangle of the map: the part being annotated or analysed. The data table export, the plot DEM metrics, Label Points from Regions and the training dataset export can be limited to it.

1. **Project ▸ Set Working Area**.
2. Click the selection button of the window, then drag a rectangle on the map; or type the top, left, width and height (in metres or in pixels).
3. **Set. Delete** removes the working area.

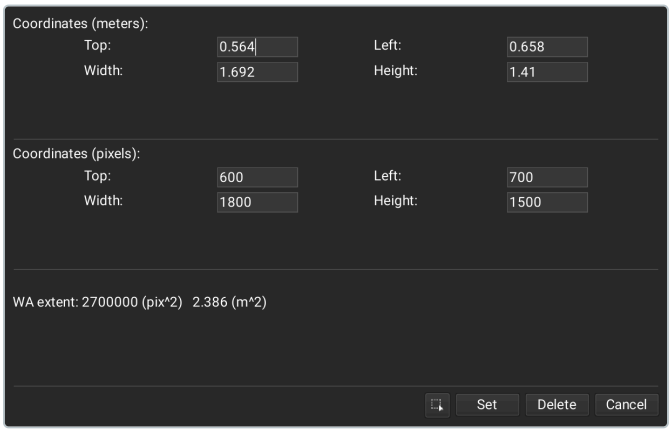


Figure 10. The Working Area window.



Figure 11. The working area drawn on the map.

3.7 Saving and autosave

- File ▸ Save Project (Ctrl + S) saves to the project file; File ▸ Save As (Ctrl + Alt + S) to a new one. TagLab first writes a temporary file and then replaces the project in one step, so a failed save never leaves a broken project file.
- **Autosave:** File ▸ Settings.. > tick **Autosave** and choose every how many minutes. It writes `<project>_autosave.json` next to the project (never the project itself), only when something changed, and waits while TagLab is busy. The setting is remembered on that computer.
- **After a crash:** open `<project>_autosave.json` with File ▸ Open Project, check it, then File ▸ Save As under the project's own name.
- When you close TagLab it asks whether to save the project.

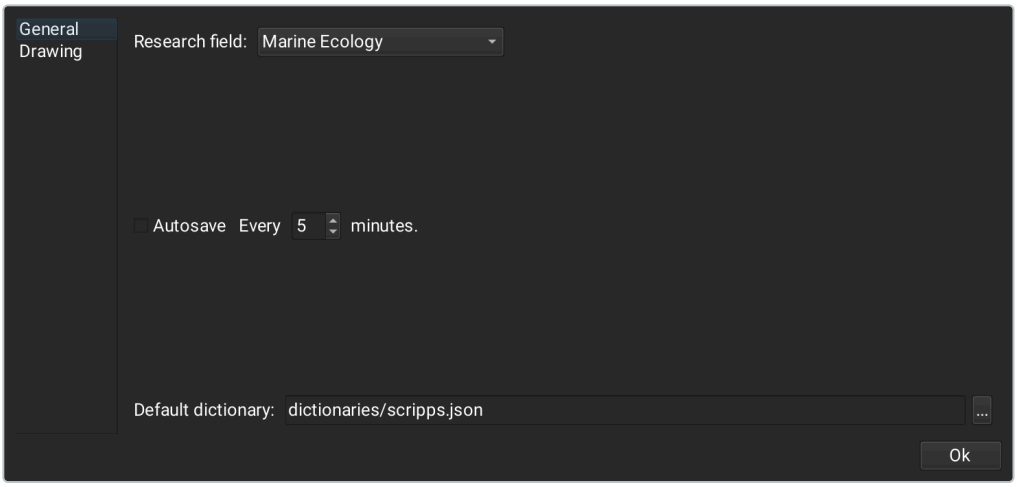


Figure 12. Settings: research field, autosave and default dictionary. The *Drawing* page sets the colours and widths of the outlines.

3.8 Moving a project to another computer

Copy the project `.json` **and** the image folders. When the project is opened on the other computer and an image is not where it was, TagLab asks you to locate it; the other images in the same folder are then found automatically. Save the project afterwards, so the new paths are kept.

4. Annotating regions

4.1 Which tool when

Situation	Best tool	Why
Most colonies, sponges, clear objects	Positive/negative clicks (RITM), 3	One click often gives the whole outline; extra clicks fix it; fast on a CPU
A compact colony with clear edges	4 clicks, 2	Four clicks on its extreme points, no confirmation needed
Low contrast, odd shapes, the AI fails	Freehand, 4	You draw exactly what you want
Fix part of an outline	Edit Border, 6, or RITM points on the selected region	Redraw only the wrong part
Two colonies drawn as one	Cut, 7	Splits a region along a line
A region in the wrong place (carry-over, nursery bricks)	Move Region	Drag, turn, then correct with points
Many small objects in an area	SAM (GPU computer)	Outlines everything in a square at once
A whole map, with a trained network	classifier, 0	See chapter 9

Every new region is created with the class **Empty**. Give it its class right away (section 4.9), or outline all the colonies of an area first and assign the classes afterwards.

4.2 Selecting regions

With the Pan/Zoom tool (1):

- **Click** a region to select it (its outline is highlighted and the Info panel shows it).
- **SHIFT + click** adds a region to the selection or removes it.
- **SHIFT + drag** draws a rectangle and selects the regions in it.
- Ctrl+A selects all, Ctrl+D none, Ctrl+I inverts the selection. The Select menu also selects by the current class, by working area, or by properties (section 6.2).
- In the tools that work on one region (Edit Border, Cut), **double-click** the region.
- **Right-click** on the map opens the region menu (section 4.11).

4.3 Positive/negative clicks (RITM)

3. An AI model draws the outline of the object under your clicks. It is the main tool of the lab.

1. **Zoom** so that the colony and a margin around it fill a good part of the view. RITM works on the area visible at the first click (the *work area*): a smaller view is faster and more precise.
2. **SHIFT + click** inside the colony (a *positive* point). The outline appears.
3. Where the outline goes **too far**, SHIFT + **right** click there (a *negative* point). Where it **misses** a part, SHIFT + click on that part. The outline is recomputed after each point. Ctrl+Z removes the last point.
4. Space confirms the region; Esc cancels.

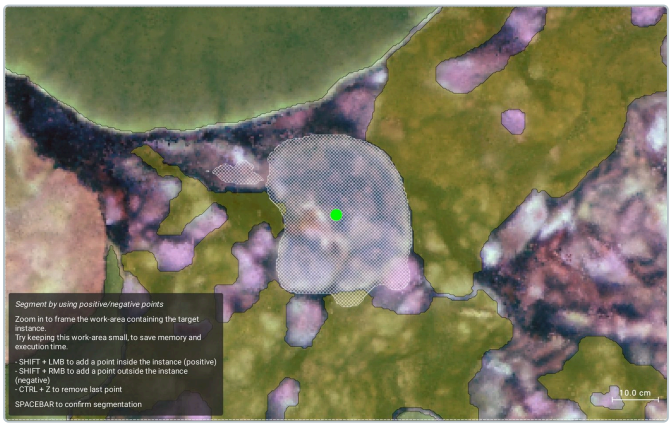


Figure 13. One positive point (green): the outline already covers the colony.

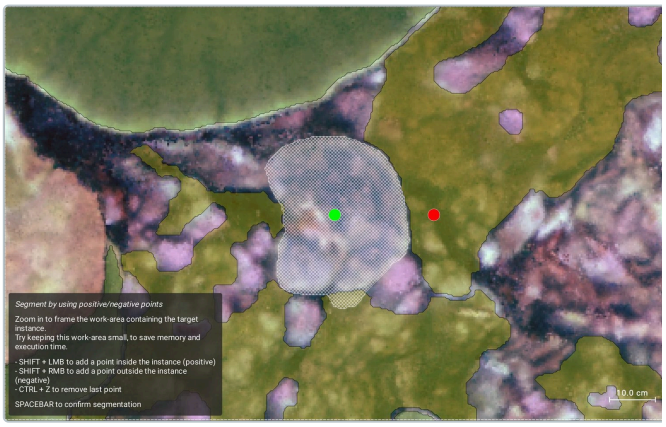


Figure 14. A negative point (red) outside stops the outline from leaking.

Correcting an existing region with RITM: select the region (Pan tool, click), choose RITM, and add points: the first point starts from the region's own outline. Space replaces the outline; the region keeps its id, class, name, note, attributes and matches (its surface area and DEM metrics are cleared, because the outline changed).

> **Good clicks**
> Put the first point well inside the colony, not on its edge. Add negative points on the substrate the outline leaked into, close to the true edge. Two or three points are usually enough; if the outline keeps jumping, press Esc and start again zoomed closer.

4.4 4 clicks (DEXTR)

2. SHIFT + click on the **top, bottom, left and right** extreme points of the colony, in any order. After the fourth click the outline is computed and added at once (no Space). Ctrl + Z removes the last point.

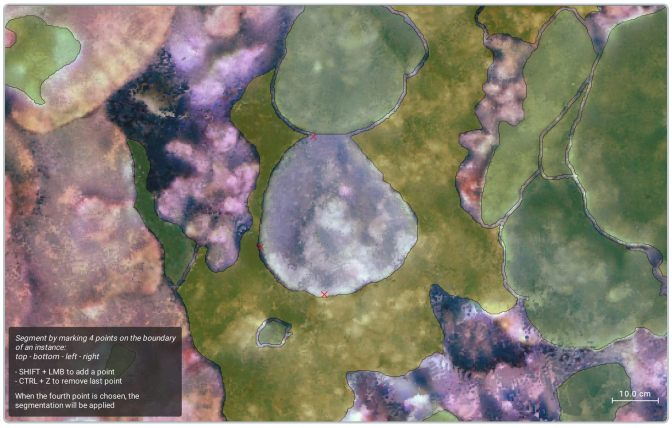


Figure 15. Three of the four extreme points placed.

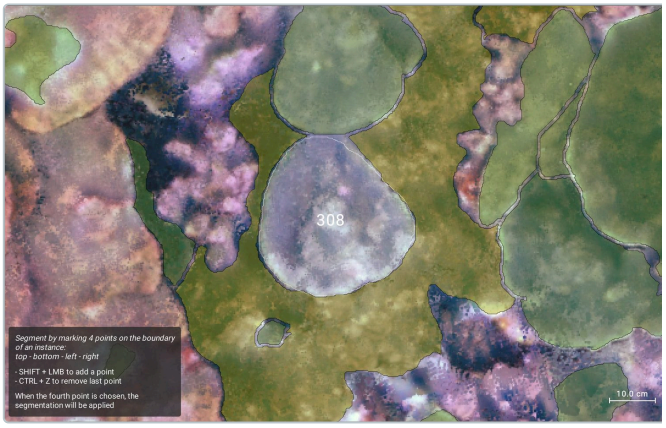


Figure 16. The fourth point gives the outline (a new region, class Empty).

4.5 Freehand

4. SHIFT + drag to draw the outline as a **closed** curve (end where you started), then Space. You can draw it in several strokes; the shape is closed between the end and the start. Zoom in for precision: the drawn line is the outline.

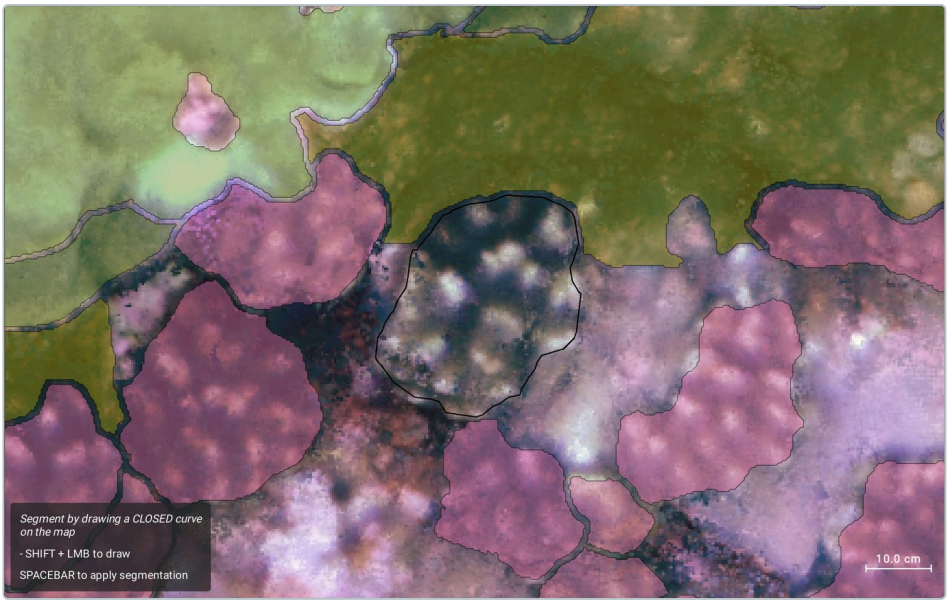


Figure 17. Freehand: the line drawn with SHIFT + drag, before Space.

4.6 SAM (Segment Anything)

SAM - all instances in an area outlines every object in a square:

1. **Step 1:** SHIFT + mouse wheel changes the size of the square; SHIFT + click places it.
2. **Step 2:** SHIFT + wheel changes the spacing of the seed points; Space runs SAM.

SAM - positive/negative clicks in an area works like RITM inside the square.

> **SAM needs the GPU computer**
> SAM is a very large model. On the annotation computers (CPU only) one square can take minutes. Check every outline it makes: SAM also outlines substrate, shadows and parts of colonies.



Figure 18. SAM, step 1: the dashed square is the work area.

4.7 Edit Border

6. Redraws part of the border of one region:

1. Double-click the region to select it.
2. SHIFT + drag a line that **starts inside the region, goes out along the correct border and comes back inside** (to add a part), or that starts outside, cuts in and leaves again (to remove a part). The line must cross the border at both ends.
3. Space applies the change.

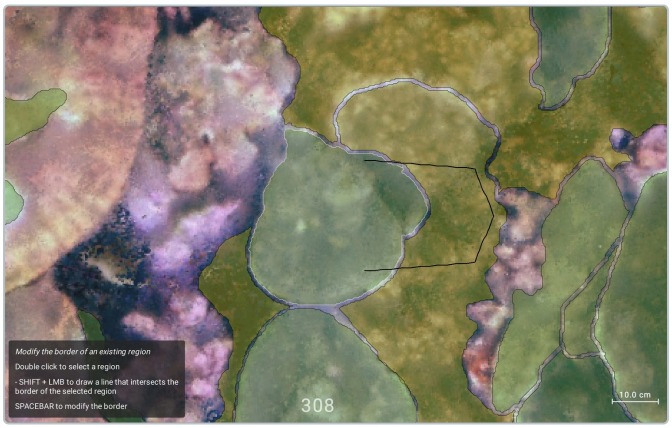


Figure 19. The line drawn from inside the region, around the missing part, and back.

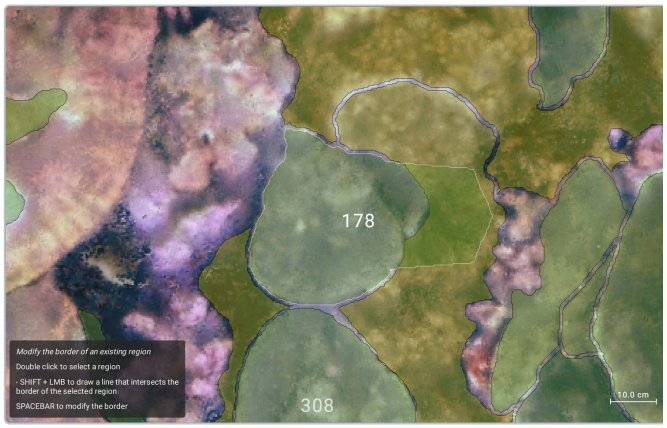


Figure 20. After Space: the part is added to the region.

4.8 Cut

7. Double-click a region, SHIFT + drag a line across it (from outside to outside), then Space: the region becomes two regions with new ids. In year-to-year projects both pieces keep the match of the original colony.

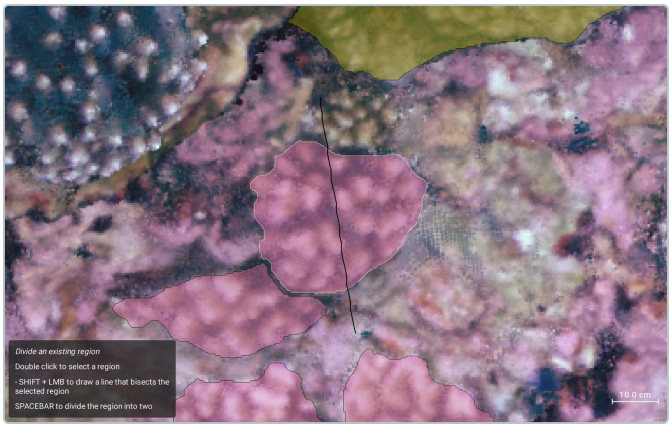


Figure 21. The cut line across the selected region.

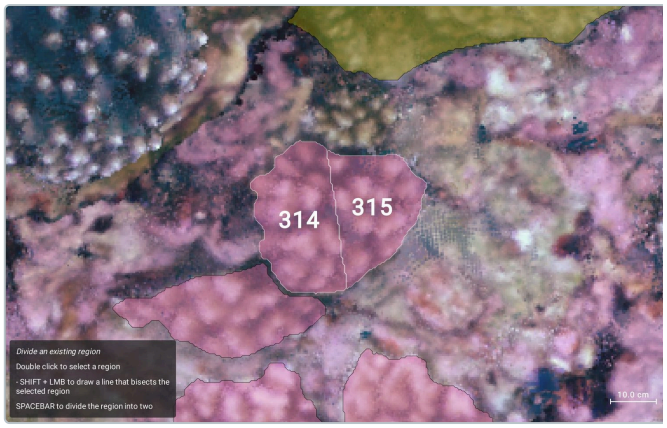


Figure 22. After Space: two regions (314 and 315).

4.9 Assigning classes

Three ways:

- **Assign tool** (5): click the class in the Labels panel, then SHIFT + click each region.
- **Several regions at once:** select them (Pan tool), then **double-click** the class in the Labels panel.
- **Key** A (Regions ▸ Assign Class to Region) gives the class selected in the Labels panel to the selected regions.

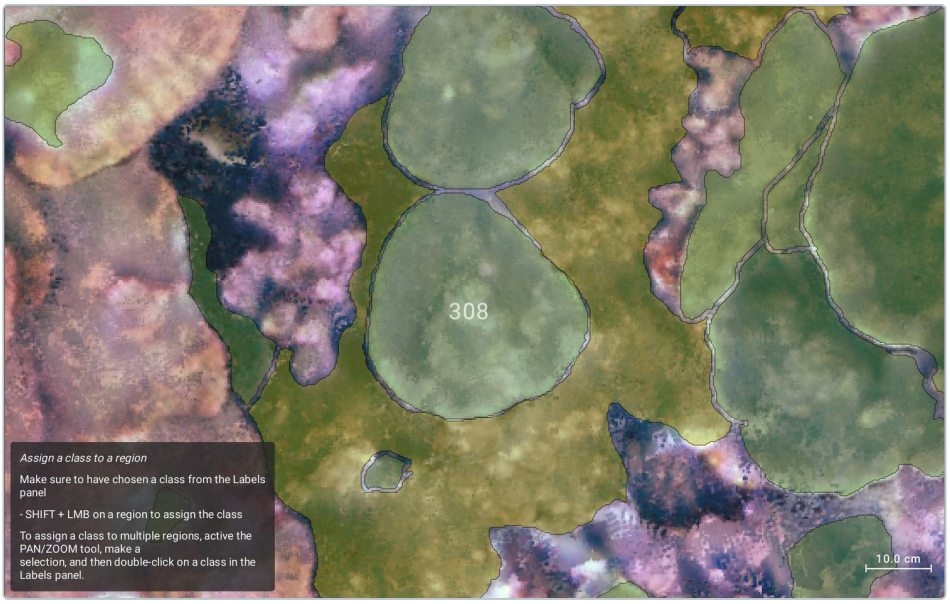


Figure 23. The region drawn with 4 clicks (class Empty, white) after SHIFT + click with the Assign tool: it now has the colony's class and colour.

A class change spreads to the same colony in the other years of the project (through the matches), except a change to Empty, which stays in its year.

4.10 Move Region

Moves and turns regions that are in the wrong place, and corrects their outline, without changing their id (so their matches and attributes stay). It is used after Carry Regions Over (chapter 8) and for nursery bricks that were moved.

Action	How
Move a region (or the whole selection)	SHIFT + drag it
Move by 1 pixel / 10 pixels	Arrow keys / SHIFT + arrow keys
Turn by 5° / 90°	, and . / SHIFT + , and . (or < and >)
Correct the outline	SHIFT + click (positive point), SHIFT + right click (negative point); Space confirms, Esc cancels, Ctrl+Z removes the last point
Re-fit the outline to the image at its new place	Space without points

Move and turn first, then add points: moving is off while there are points. A region moves by whole pixels, so its area does not change; surface area and DEM metrics are cleared (they were measured at the old place).

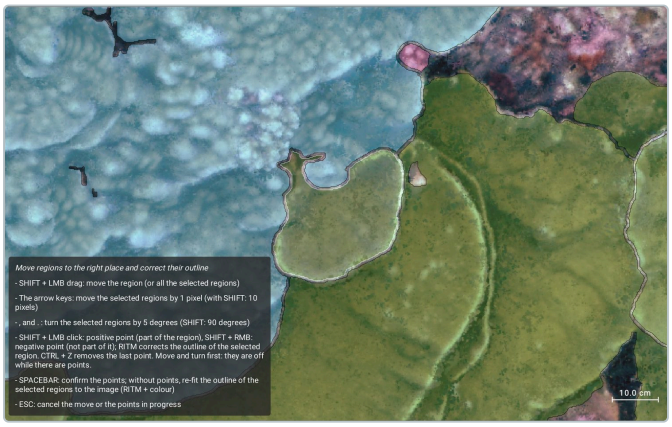


Figure 24. A selected region with the Move Region tool.

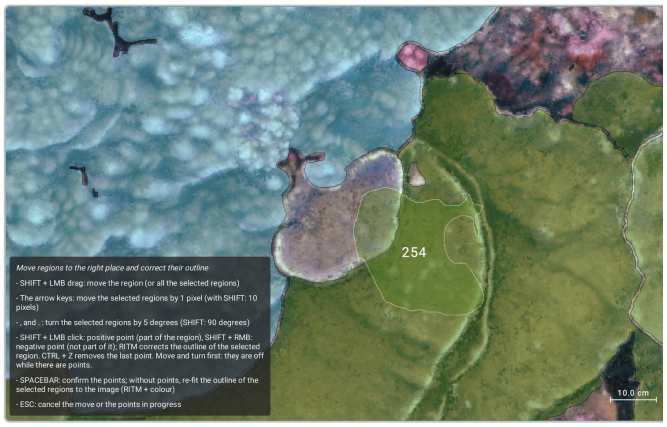


Figure 25. After SHIFT + drag and a 90° turn (SHIFT + .).

4.11 Region operations (Regions menu and right click)

Select the regions, then use the key, the **Regions** menu or the right-click menu. Operations that need two regions are greyed out until two are selected.

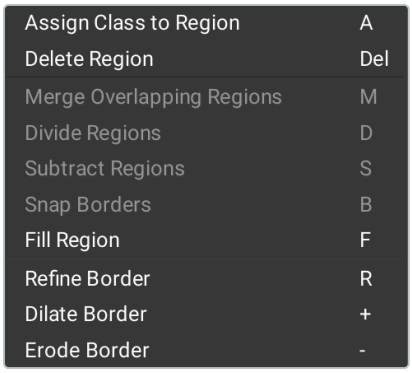


Figure 26. The right-click menu on a selected region.

Operation	Key	Selection	Result
Assign Class	A	any	The active class of the Labels panel
Delete	Del	any	Removes the regions
Merge Overlapping Regions	M	2 or more	One region, the union
Divide Regions	D	2 overlapping	The overlap is shared out, so they no longer overlap
Subtract Regions	S	2	The first selected minus the second
Snap Borders	B	2 touching	Their common border is made identical (no gap, no overlap)
Fill Region	F	any	Holes filled
Refine Border	R	1	Border moved onto the nearby image edges
Dilate / Erode	+ / -	any	Grown / shrunk by about one pixel

- **Regions ▸ Refine All Borders** refines every region of the map.
- **Regions ▸ Create a Background Region using the WA** (N) creates Empty regions over every part of the working area that has no region (for example to class the rest as Sand).

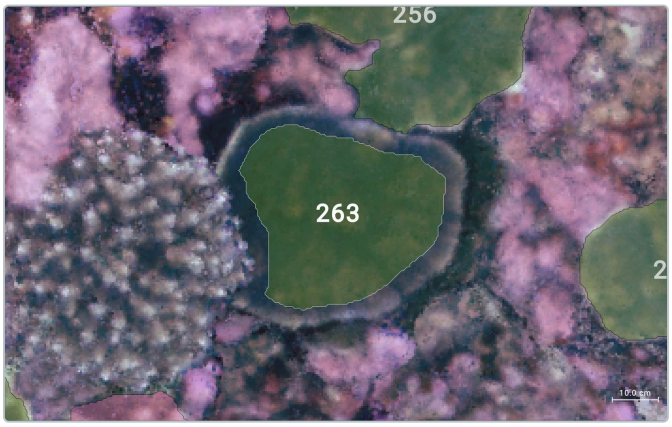


Figure 27. A region drawn too small (region 263).

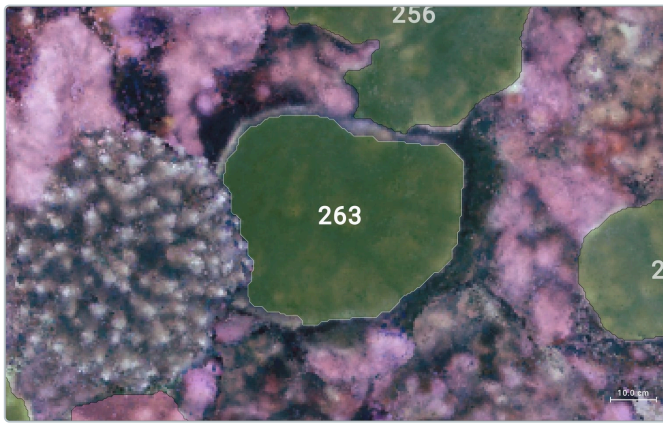


Figure 28. After Refine Border (R): the border follows the colony edge.

4.12 Undo and redo

Ctrl + Z undoes the last operation and Ctrl + Shift + Z redoes it. Each viewer (each map in split screen) has its own history. The comparison tables follow undo and redo. Operations on **points** cannot be undone.

4.13 Class, note and attributes of a region

The **Info and Attributes** panel shows the selected region:

- **Properties:** id, genet, class, area (cm^2), perimeter (cm) and surface area (cm^2, 0.0 until it is computed).
- **Attributes:** the region attributes of the project (section 3.5) and the values TagLab adds (DEM metrics, carry-over status); type the values here. Below them, the **Notes** field (free text, saved with the region and exported).
- **AI Suggestions:** class suggestions imported from CoralNet for points.

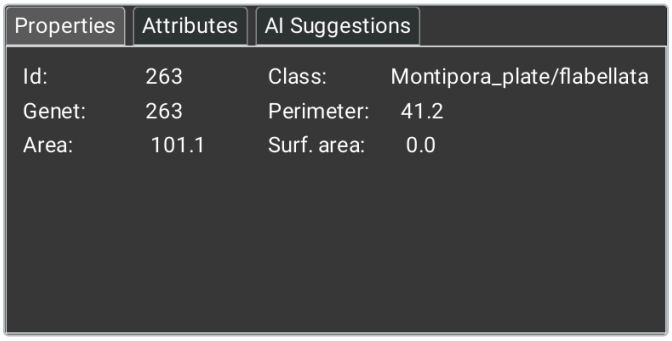


Figure 29. Properties of the selected region.



Figure 30. The Attributes tab of a project without region attributes: only the Notes field.

4.14 Other tools

- **Place annotation point** : see chapter 5.
- **Create crack** (8, no button): draws a thin gap through a region along a line.
- **Watershed** and **Rows** are for the *Digital Heritage* research field and are hidden in the lab's setting.
- **Grid** : divides a large map into cells to annotate one at a time. Create it (width, height, rows, columns, position), then turn grid operations on: the right-click menu marks the cell under the mouse as empty, incomplete or complete and adds a note; U changes the state of the cell under the mouse; G shows or hides the grid.

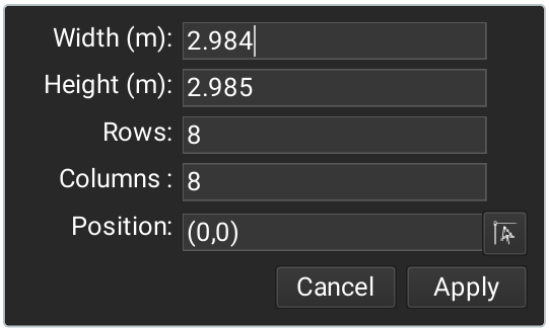


Figure 31. Create grid: its size in metres, rows and columns, and its top-left corner (the button picks it on the map).

4.15 SOP: annotating a plot

1. Open the project and choose the map in the **Map** list. Check its pixel size in the Maps Editor.
2. If only part of the map is to be annotated, check that the working area is set.
3. Work systematically, for example in horizontal bands from top left, so that no colony is missed. Turn on the grid for large maps.
4. Outline each colony with RITM (or another tool, section 4.1) and give it its class at once.
5. Every 10–15 minutes, or after a difficult colony, save (Ctrl + S).
6. When the area is done, check: turn the fill off (Q) to see the edges, look for Empty regions (**Select ▸ Select by current Class** with Empty active), and sort the Data table by area to find tiny regions made by mistake.
7. Save, and note in the lab log which part of the map you annotated.

5. Points

Points are single annotated pixels, as in a point-count survey. Each point has a class (Empty when new) and can have a note. Points are used for percent-cover estimates and to make training data for CoralNet (chapter 10).

> **Points have no undo**
> No point operation can be undone with Ctrl+Z. Save the project before you sample, label or import points.

5.1 Placing points by hand

Place annotation point: SHIFT + click on the map adds a point (class Empty). Then:

- **Select** points like regions with the Pan tool (click, SHIFT + click, SHIFT + drag).
- **Give a class:** select the points, click the class in the Labels panel and press A, or double-click the class.
- **Delete:** select and press Del.
- The Data table lists the points (the *Regions* and *Points* boxes above it choose what it shows) and the Labels panel counts them (#P).

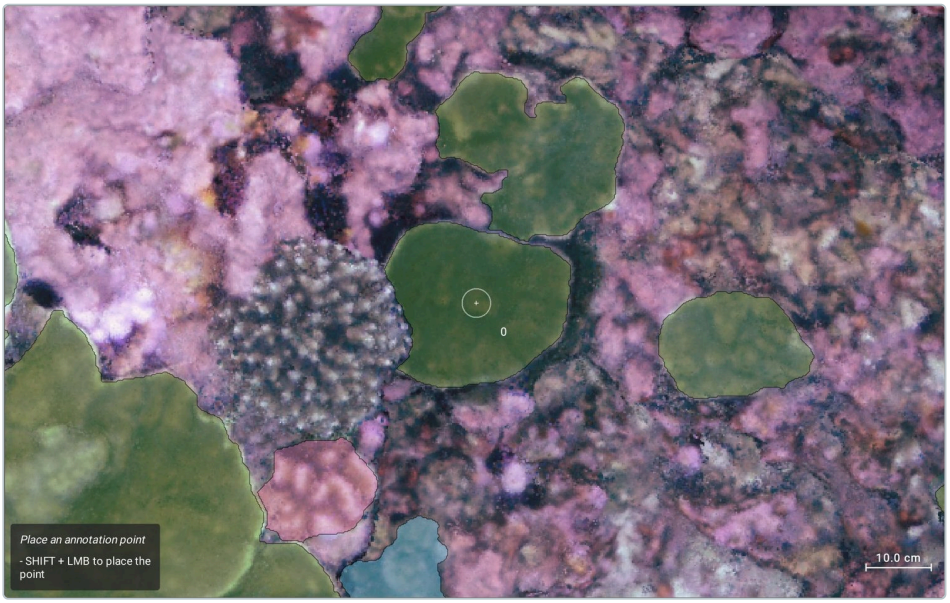


Figure 32. Placing points: each point is drawn in the colour of its class.

5.2 Sampling points on a map

Points ▸ Sample Points On This Map places points automatically, inside one or more *sampling areas*:

1. Choose where the sampling areas go:
   - **Add a single sampling area:** click the selection button, then click or drag on the map to place it (or type its top and left corner in pixels).
   - **Sampling the Working Area with multiple random areas:** a number of areas placed at random inside the working area.
   - **Sampling a transect with multiple sampling areas:** click the line button, SHIFT + click the two ends of the transect, then choose the number of areas and *Equi-spaced* or *Random*.
2. Set the sampling area itself: **Method** (*Grid*: a regular grid; *Random*; *Stratified*: one random point in each cell of a grid), **# Points**, an **Offset** (a margin left free inside the area) and the **Width** and **Height** of the area (in px or cm).

3 **Generate.** The points are added as Empty; the sampling areas are drawn on the map (layer *Sampling areas*).

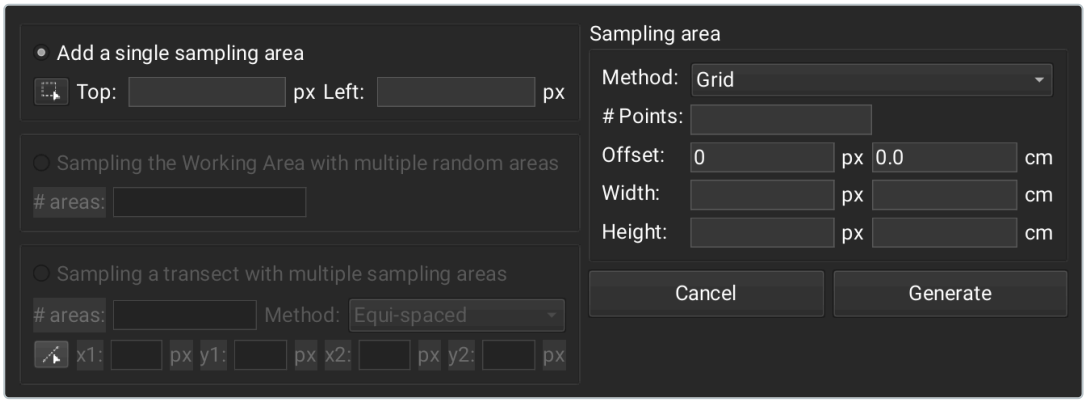


Figure 33. Sample Points On This Map.

5.3 Labelling points from the regions

When the map is already outlined, **Points ▸ Label Points from Regions...** gives each point the class of the region under it, and adds new points inside the regions so that every class has enough training points for CoralNet. It works on the active map, inside the working area if one is set.

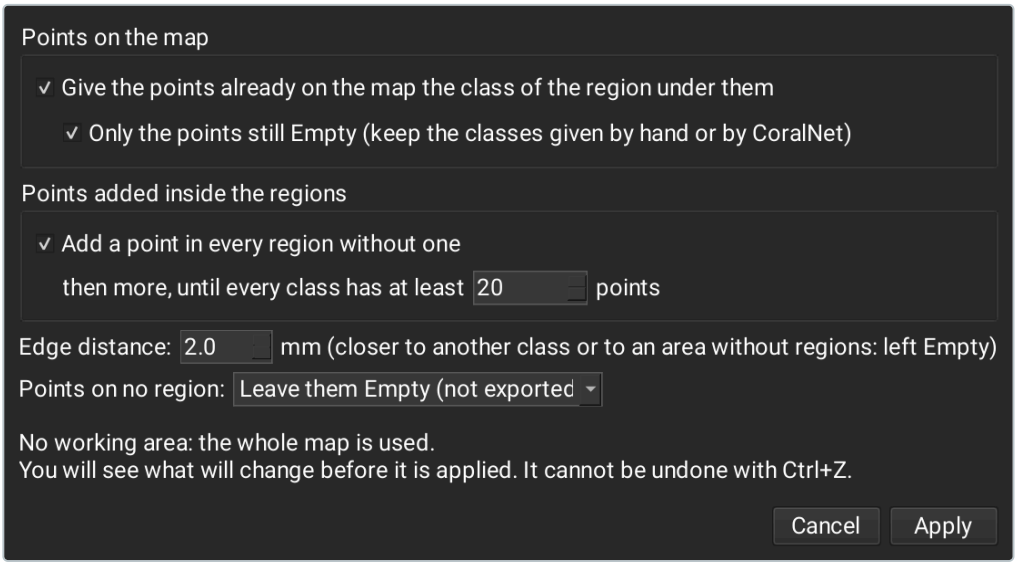


Figure 34. Label Points from Regions.

Option	Meaning
Give the points already on the map the class of the region under them	Labels existing points (for example from Sample Points)
Only the points still Empty	Keeps the classes given by hand or imported from CoralNet
Add a point in every region without one, then more until every class has at least *N* points	New points, spread over the regions of each class by area (default 20)
Edge distance (default 2 mm)	A point closer than this to another class, or to an area without regions, stays Empty and is not exported: its label is uncertain. New points keep this distance from edges and twice it from each other
Points on no region	Leave them Empty, or give them a class (for example Sand, when the sand is not outlined)

Where regions overlap, the smaller one wins (for example a sponge drawn on a sand region without a hole). Before anything changes, a summary shows what will happen; **Yes** applies it. Running it again with the same options gives the same points.

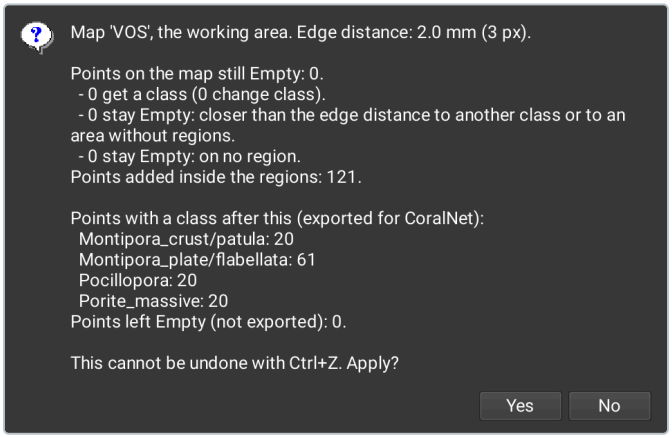


Figure 35. The summary shown before the points are changed.

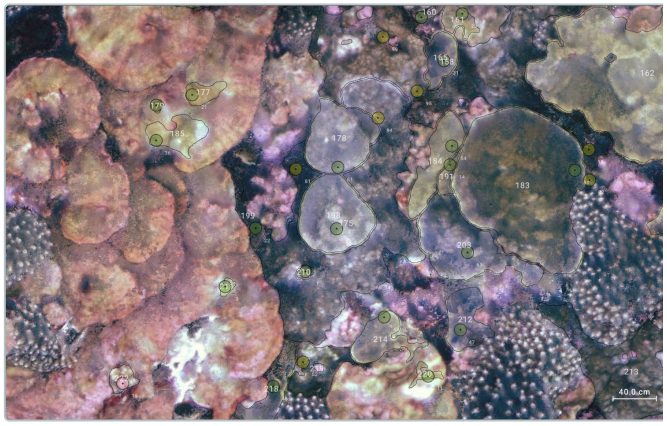


Figure 36. The points added inside the regions (fill turned off with Q).

5.4 Importing points

- **Points ▸ Import CoralNet Point Annotations**: a CoralNet annotation CSV (for example the confirmed annotations or the machine suggestions of a source), with at least the columns Name, Row and Column. Points on the tiles exported by TagLab are put back at their place on the map. Classes not in the project are added to the labels (give them a colour in the Labels Dictionary Editor). CoralNet's machine suggestions and confidences appear in the *AI Suggestions* tab of the Info panel.
- **Points ▸ Import Viscore Point Annotations**: a Viscore points CSV; the window asks for the scale.
- **Points ▸ Export CoralNet Point Annotations**: all the points of the map (or of the working area) as one CoralNet CSV.

For training a CoralNet source with your regions, see chapter 10.

6. Checking and exporting

6.1 The Data table

The **Data table** panel lists the regions (Type R) and points (Type P) of the map shown, with their id, class and area (cm^2).

- Click a row to select the region and show it on the map; Ctrl or SHIFT + click selects several rows.
- Click a column header to sort, for example by **Area** to find tiny regions made by mistake, or by **Class**.
- Type an id in **Search Id** to find a region.
- The **Regions** and **Points** boxes choose what is listed.

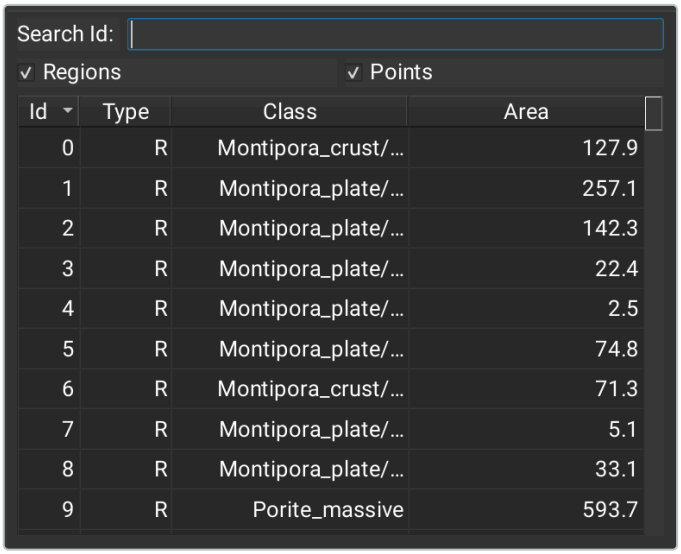


Figure 37. The Data table.

6.2 Selecting by properties

Select ▸ Select by Properties selects the regions whose area, perimeter, class or position (of the centroid or of the whole region) are within the ranges you tick. **Add** adds them to the selection, **Remove** takes them out; **All**, **None** and **Invert** work on the whole map. The bottom line gives the number of regions and of selected ones. Use it to check, for example, all regions smaller than 1 cm^2.

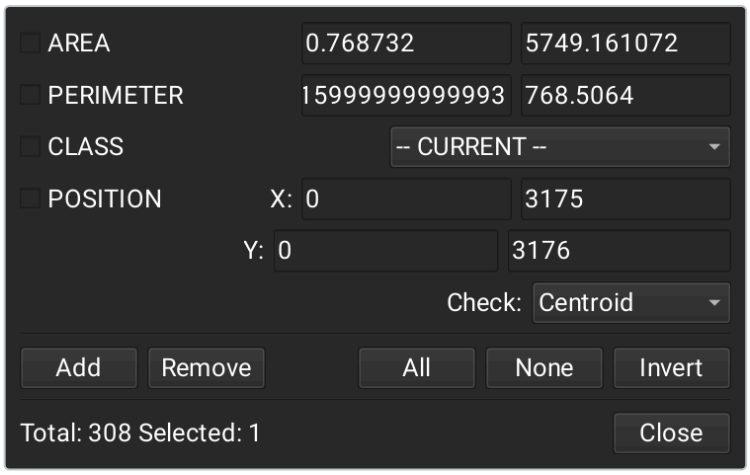


Figure 38. Select by Properties: area and perimeter ranges, class, position.

6.3 Shape measurements

Regions ▸ Compute Geometric Info (with regions selected) measures each selected region: area and perimeter, the area of its convex hull and bounding box, the sides and orientation of the smallest rectangle around it, and the axes and orientation of a fitted ellipse (mm and mm^2). It can colour the regions by a measure, draw the rectangles or ellipses, **Export to CSV**, or **Add Geometric Measurements as Region Attributes** (then they are exported with the data table).

6.4 Exports

All exports are in **File ▸ Export**, except the ones named otherwise below. They export the **map shown** (the active map in split screen).

Export	What you get	Needs
File ▸ Export ▸ Export Annotations As Data Table	A CSV, one row per region and/or point (the window asks: Regions, Points or Both). Columns in Appendix D	Only visible classes; only inside the working area, if one is set
File ▸ Export ▸ Export Regions As Shapefile	The outlines as GIS polygons, with the same columns as the CSV (attribute names are cut to 10 characters)	A georeferenced map
File ▸ Export ▸ Export Regions As Labeled Image (PNG)	An image of the map's size, each region painted in its class colour	—
File ▸ Export ▸ Export Regions As A GeoTiff	The same label image as a georeferenced GeoTIFF (working area only, if set)	A georeferenced map
File ▸ Export ▸ Export Orthoimage As A GeoTiff	The map image as a georeferenced GeoTIFF (working area only, if set)	A georeferenced map
DEM ▸ Export Clipped Raster	The DEM cut to the regions (or to those of the working area)	A DEM
File ▸ Export ▸ Export Histogram From Regions	A histogram of the colony sizes of the classes you tick (Preview, Save As)	—
File ▸ Export ▸ Export Regions As DXF / **File ▸ Export ▸ Export Regions As SVG**	Outlines for CAD or drawing programs	Their georeferenced option does not work in this version
DEM ▸ Export Plot Metrics (CSV)	One row per map with the plot DEM metrics	DEM metrics computed (chapter 7)
Compare ▸ Export Matches, **Compare ▸ Export Genet Data as CSV**	Year-to-year tables	Chapter 8
Train ▸ Export New Training Dataset	Tiles for training a network	Chapter 9
Points ▸ Export Tiled Data for CoralNet	Tiles and point labels for CoralNet	Chapter 10

> **Georeferenced map**
> A map is *georeferenced* when its orthomosaic is a GeoTIFF with coordinates (as exported by Metashape or Agisoft with a coordinate system). TagLab then knows where each pixel is on Earth, and the shapefile and GeoTIFF exports are possible. Plain `.png` or `.jpg` maps are not georeferenced. The Maps Editor shows which maps are.

Before you export

Check the pixel size of each map (Maps Editor), compute surface areas and DEM metrics first if you need them in the table, and make sure every class you need is visible (the eye in the Labels panel): hidden classes are left out of the data table.

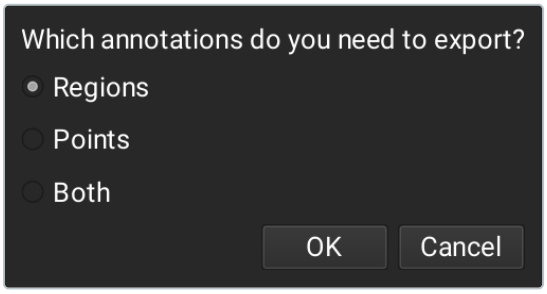


Figure 39. Export Annotations As Data Table asks what to export.

Importing (File ▸ Import):

- **Import Label Image:** turns a label image of the map's size (one colour per class, for example from another program) into regions. Colours are matched to the project's labels.
- **Import Shapefile** (georeferenced map): as a *Labeling layer* its polygons become regions (choose the field that holds the class and the fields to keep); as a *Reference layer* or *Other* it is only drawn (Layers panel).
- **Add Another Project:** adds the maps of another project, with their annotations, to this one.

7. DEM: surface area and 3D metrics

A **DEM** (digital elevation model) gives the height of every pixel of the map. It comes from the same photogrammetry survey as the orthomosaic and is added to the map as its *Depth Image* (section 3.1). With a DEM TagLab measures the 3D shape of each colony and of the whole plot.

> **Sample data**
> The screenshots of this chapter use the official sample *Plot18* (a reef plot from Moorea, with its DEM at 0.667 mm per pixel).

7.1 Seeing the DEM

DEM ▸ Switch RGB/DEM (C) shows the DEM in grey (dark = low, light = high) instead of the image, and back. Outlining is usually done on the image, but the DEM helps to see the edges of colonies on a busy background.

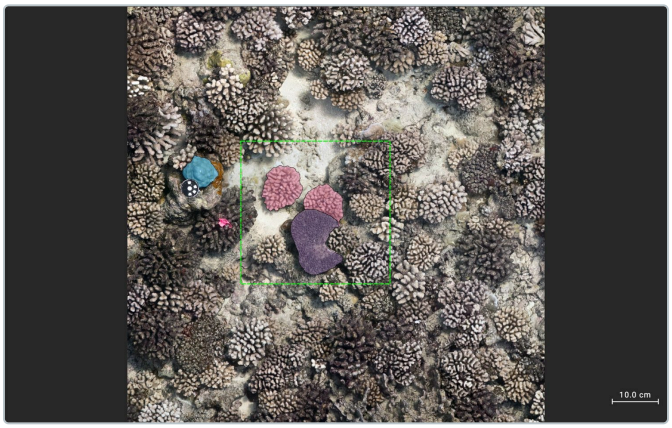


Figure 40. Plot18: the orthomosaic.

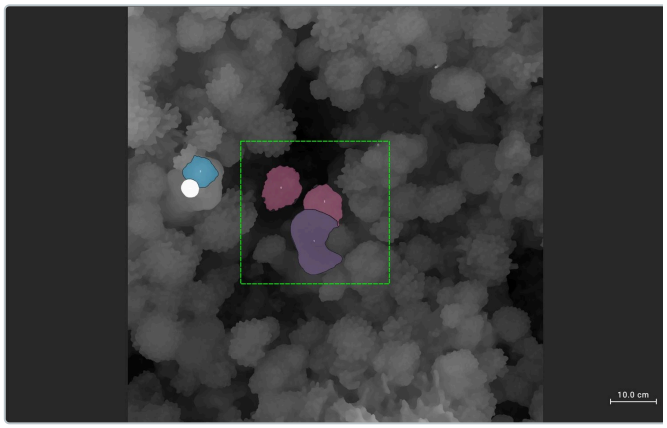


Figure 41. The same view with the DEM (C).

7.2 Surface area

DEM ▸ Calculate Surface Area computes the 3D surface area of **every region** of the map: each pixel counts as its planar area divided by the cosine of the local slope. It appears as *Surf. area* in the Info panel, the data table, the CSV exports and (when chosen) the comparison table. It is cleared when the outline of a region changes: compute it again after editing.

7.3 Colony DEM metrics

DEM ▸ Calculate DEM Metrics computes the metrics below for the **selected regions** or, with nothing selected, for **all the regions** of the map and for the whole plot (section 7.4). They are stored as region attributes (the *Attributes* tab of the Info panel) and exported as columns of the data table and of the shapefile.

Attribute	Unit	Meaning
`slope_deg`	degrees	Mean slope of the colony surface (Horn's method, as GIS programs compute it; slopes over 87°, the vertical walls, are left out)
`elev_mean`, `elev_min`, `elev_max`	m	Mean, lowest and highest elevation of the colony, in the units of the DEM
`relief_cm`	cm	Highest minus lowest point of the colony

Attribute	Unit	Meaning
`height_cm`	cm	Height of the colony's highest point above the substrate around it. The substrate is a plane fitted to a 2-pixel ring just outside the outline
`volume_cm3`	cm^3	Volume between the colony surface and that substrate plane (parts below the plane count as zero)
`rugosity`	—	Surface area / planar area of the colony (1 = flat)

- A region with less than half of its pixels covered by the DEM gets no values.
- Slopes use the DEM's own pixel size; areas and volumes use the map's pixel size. If the two differ by more than 5%, a warning says so (check the pixel size in the Maps Editor).
- The values are cleared when the outline changes (editing, moving, aligning with height correction): compute them again before exporting.

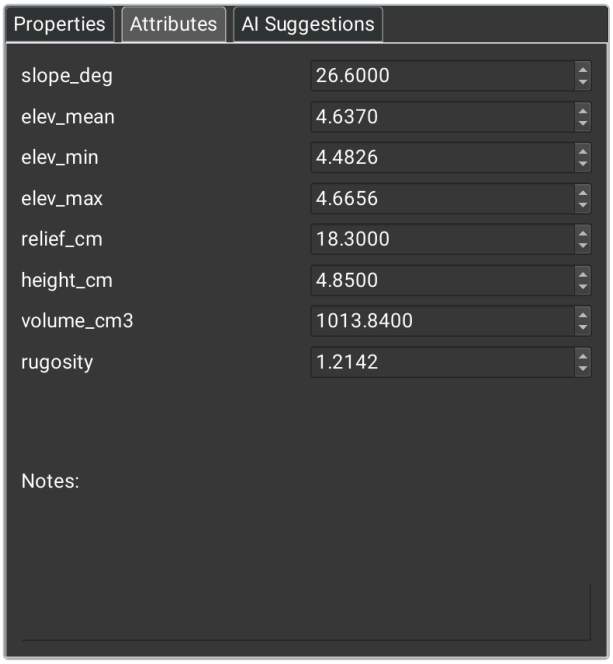


Figure 42. DEM metrics of a colony in the Attributes tab.

7.4 Plot metrics (plot rugosity)

With **no region selected**, Calculate DEM Metrics also measures the whole plot: the working area, or the whole map when there is none (only the part covered by the DEM counts). The results are shown in a message and saved with the map.

Value	Meaning
`rugosity`, `slope_deg`	3D surface area / planar area, and mean slope, at the DEM's own resolution
`rugosity_1cm`, `slope_1cm`	The same on the DEM averaged to 1 cm cells. **Use these to compare years or surveys**: a finer DEM always gives a higher rugosity
`relief_cm`	Height range of the plot after removing its overall tilt (1st to 99th percentile, so single spikes don't count)
`planar_m2`, `dem_coverage`, `dem_mm`	Area used, the fraction of it covered by the DEM, the DEM pixel size

Unlike the colony metrics, the plot values keep the very steep slopes (walls of colonies and boulders), as the usual surface-area ratio does.

DEM ▸ Export Plot Metrics (CSV) writes one row per map (columns `map, date, area, top, left, width, height, planar_m2, dem_coverage, dem_mm, rugosity, slope_deg, cell_mm, rugosity_1cm, slope_1cm, relief_cm`). Its message lists maps without values, or computed on a different area than the current working area.

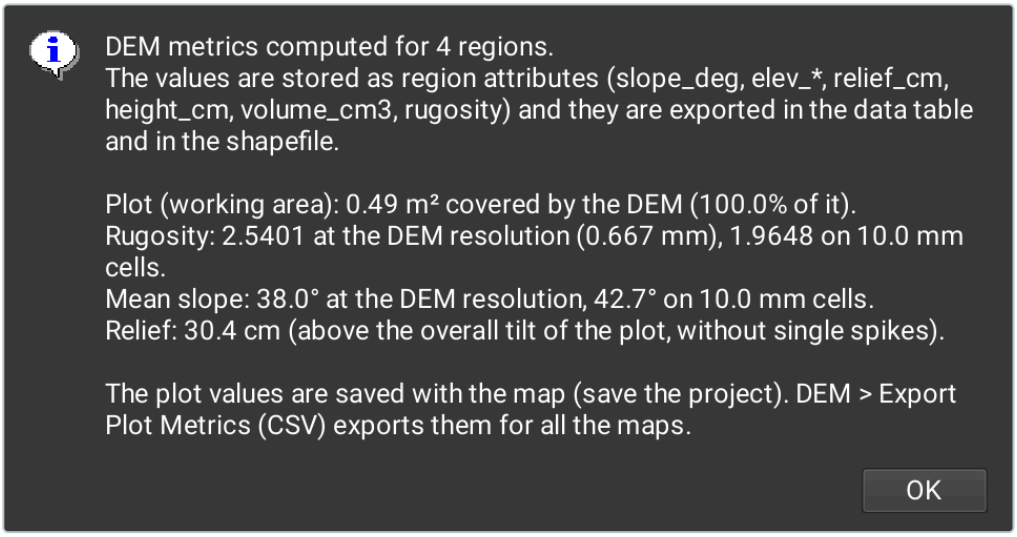


Figure 43. The message after Calculate DEM Metrics with nothing selected on Plot18.

7.5 SOP: DEM metrics of a plot

1. Check that the map has a DEM (Maps Editor) and that its pixel size is right.
2. Set the working area to the plot (section 3.6), so that the plot values use the same area every year.
3. Finish and check the outlines first: editing a region clears its values.
4. **Select ▸ Select None** (Ctrl + D), then **DEM ▸ Calculate DEM Metrics**. Read the message (coverage, warnings).
5. **DEM ▸ Calculate Surface Area** if you need surface areas.
6. Save the project, then export the data table and **DEM ▸ Export Plot Metrics (CSV)**.

8. Year-to-year comparison

When a plot is surveyed again, its new map is added to the same project and TagLab follows each colony from one survey to the next. Each colony of the older map is **matched** to the same colony in the newer one; from the two outlines TagLab computes its **fate** (grew, shrank, stayed the same, died, was born, split or fused). The chain of matches through all the surveys is the colony's **genet**.

> **Sample data**
> The screenshots use the official 2013 and 2014 maps of the same reef plot. For the Alignment Tool, a copy of the 2014 map was turned by 7° and shifted, to look like a new survey that is not yet aligned.

8.1 The workflow

1. **Add the new survey** to the plot's project (**Project ▸ Add New Map...**, section 3.1).
2. **Align it** to the previous survey with the Alignment Tool (8.2), so that the same colony is at the same pixel position in both maps. Skip this only if the maps were made in the same coordinate system and line up already.
3. **Open split screen** with the older map on the left and the newer one on the right (8.3).
4. **Annotate the new map**, in one of two ways:
   - **Carry Regions Over** (8.5): TagLab copies the old outlines onto the new map, fits them to the new image and flags the doubtful ones for review. Then add the recruits by hand. Fastest for most plots.
   - Or outline the new map from scratch (chapter 4), then **Compute Automatic Matches** (8.4).
5. **Check the comparison table** (8.6): the colonies to check, the dead and born ones, wrong matches.
6. **Export** the matches and the genets (8.7). Save the project.

8.2 Aligning a new survey: the Alignment Tool

Project ▸ Alignment Tool lines up a map (*Image to align*) on another one (*Reference Image*) with matching points called **markers**. It can turn, shift and (optionally) scale the map, then writes a new, aligned copy of its files. The original files are never changed.

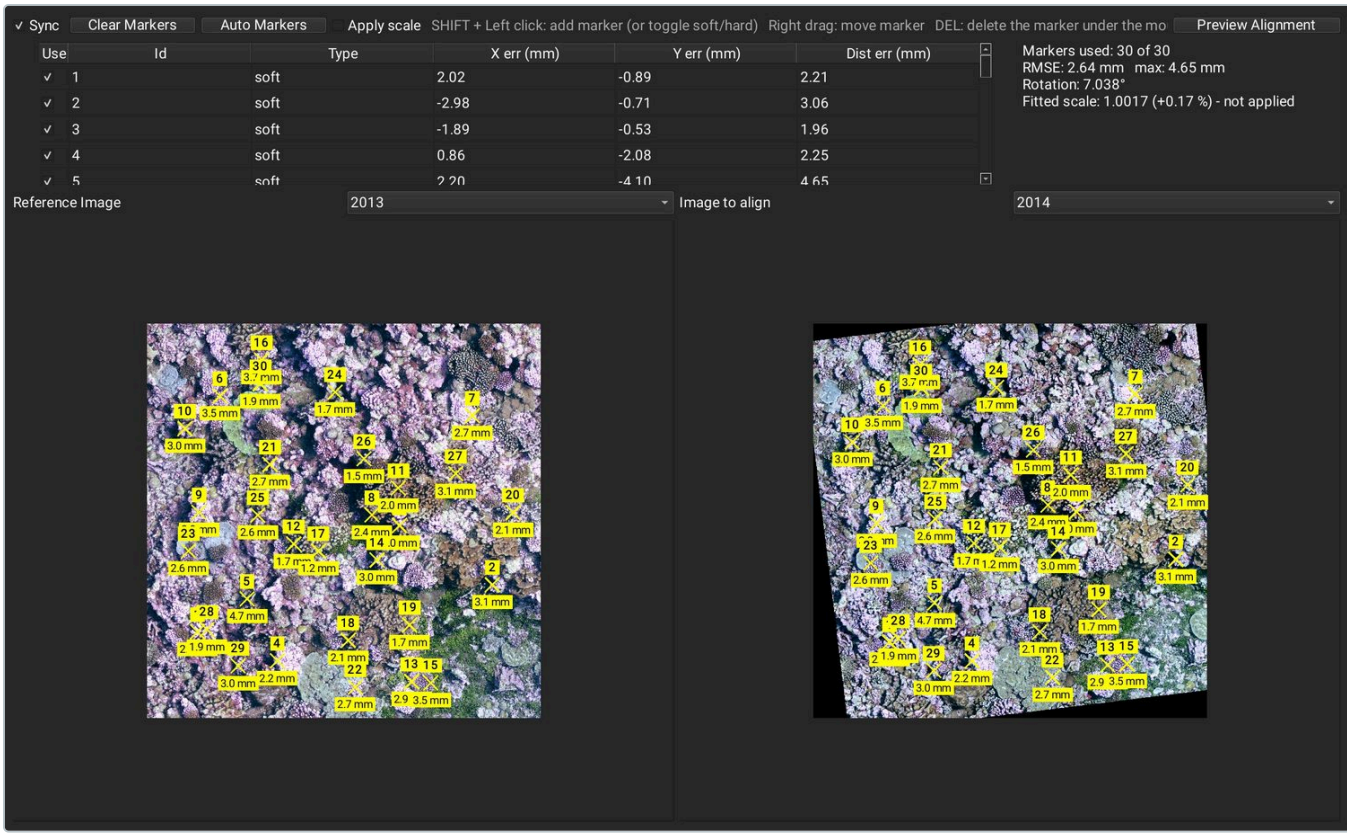


Figure 44. The Alignment Tool after Auto Markers: the reference (2013, left) and the map to align (the turned 2014 copy, right), with the markers in both.

Markers

- **Auto Markers** finds them by itself, whatever the rotation between the maps (it took a few seconds here and found 30 markers). With DEMs on both maps, *Match on* can use the **DEM relief** instead of the colours: it still works after bleaching, algae growth or a change of light.
- **By hand:** SHIFT + click a recognisable spot on the left map, then SHIFT + click the same spot on the right map. SHIFT + click on a marker switches it between *soft* (yellow) and *hard* (red, three times the weight). Drag a marker with the **right** button to move it; Del deletes the marker under the mouse; **Clear Markers** deletes them all.
- Use at least 4–6 markers spread over the whole plot, on things that don't move (rock, dead colony skeletons, the plot pins), not on soft corals, sponges or sand.
- **Sync** moves and zooms both maps together.

Use	Id	Type	X err (mm)	Y err (mm)	Dist err (mm)
✓	1	soft	2.02	-0.89	2.21
✓	2	soft	-2.98	-0.71	3.06
✓	3	soft	-1.89	-0.53	1.96
✓	4	soft	0.86	-2.08	2.25
✓	5	soft	2.20	-4.10	4.65

Figure 45. The marker table: untick *Use* to leave a marker out. X err, Y err and Dist err are how far each marker is from where the fit puts it (mm).

The fit is shown on the right of the table:

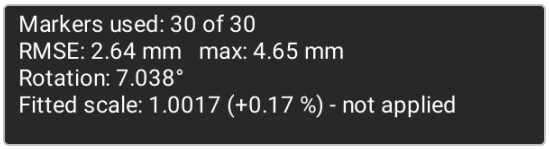


Figure 46. Fit summary: markers used, RMSE (the typical marker error) and the largest error, the rotation, and the scale difference (applied only with *Apply scale*).

> **Reading the errors**
>
> A good alignment has an RMSE of a few mm and no marker much worse than the others. A marker with a large error (red in the table) is usually on something that moved or was clicked at the wrong spot: untick it or delete it. Leave **Apply scale** off unless the two surveys clearly have different scales (the fitted scale is far from 1).

Checking the alignment

Preview Alignment shows the two maps on top of each other. Each pane has its own view:

View	What it shows
Blend	The two maps mixed (slider: how much of each)
Red / cyan overlay	Reference in red, aligned map in cyan. Grey where they match; a shift shows as red and cyan edges side by side
Difference	Bright where the maps differ
Checkerboard	Squares of one map and of the other: edges must continue across the squares
Swipe	Reference left of a line that follows the mouse, aligned map right of it
Flicker	The two maps in turn: a shift shows as a jump
DEM shading / Height difference	With DEMs on both maps (see below)

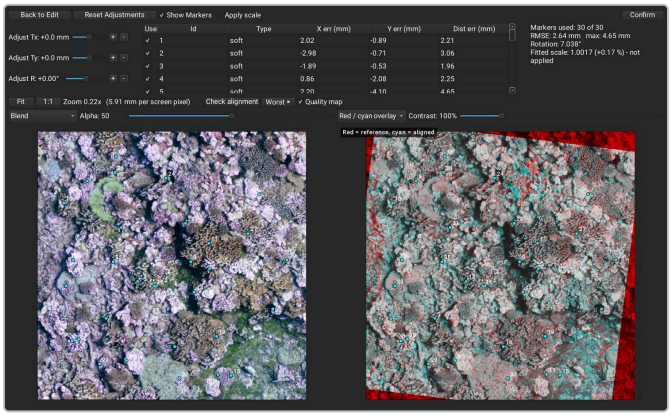


Figure 47. Preview: Blend (left) and red / cyan overlay (right).

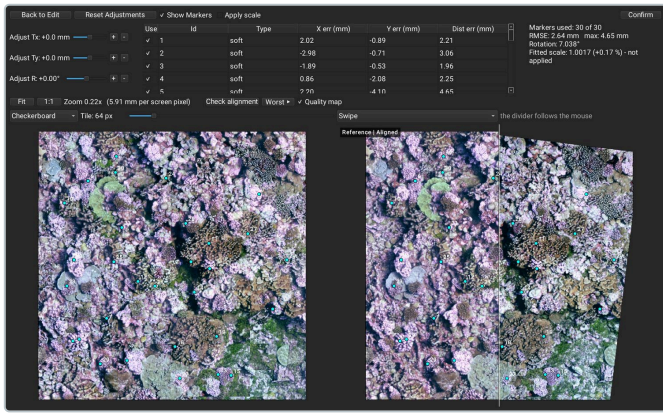


Figure 48. Checkerboard (left) and Swipe (right).

Fit and **1:1** set the zoom; **Adjust Tx / Ty / R** move or turn the aligned map by hand, in small steps (**Reset Adjustments** removes them). **Back to Edit** returns to the markers.

Check alignment measures what is left of the misalignment on a grid of about 12 × 12 cells. Each cell is green (≤ 1 mm), yellow (≤ 3 mm), red (> 3 mm) or grey (not measurable: no texture, no data, or the place changed), with an arrow showing the direction. **Worst ▶** zooms on the worst cell, then the next one.

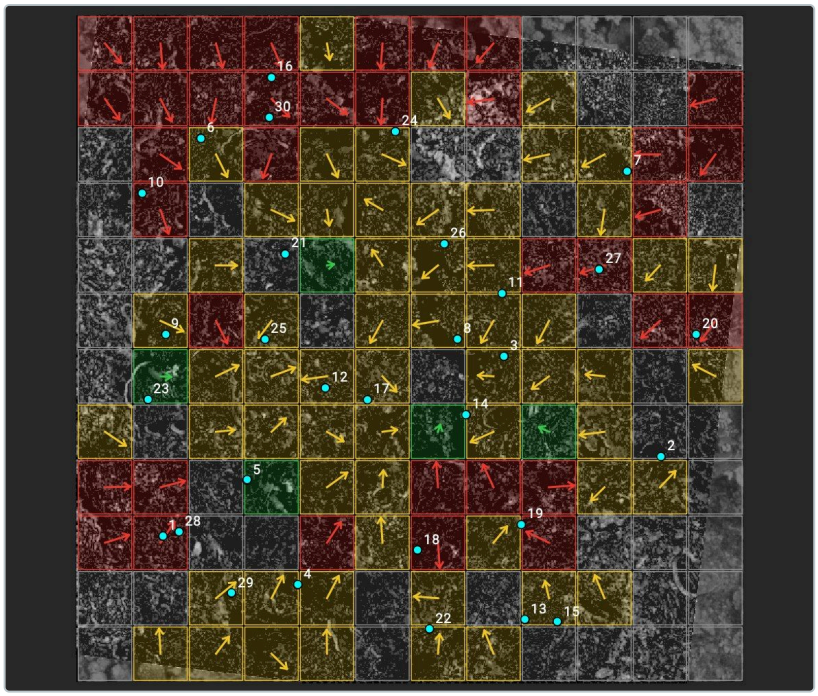


Figure 49. The quality map on the aligned 2014 copy.

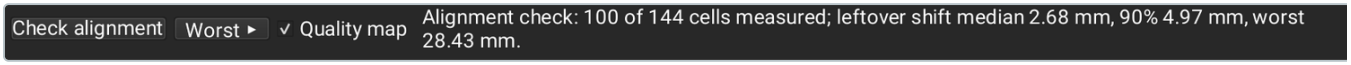


Figure 50. The result line: median, 90% and worst leftover shift.

In this example most cells are yellow (median 2.7 mm): 2013 and 2014 are two separate photogrammetry models with small local distortions, and the colonies grew in between. Red cells are mostly places that changed. If most arrows point the same way, the label says how far the map sits off; fix it with more markers or with Adjust Tx / Ty.

With DEMs on both maps:

- *Show: DEM shading* shades both DEMs with the same light; every view and the check can use it.
- TagLab compares the heights and fits the difference with a tilted plane (height offset in cm, tilt in degrees). **Match DEM heights to the reference** (on by default) removes that offset and tilt from the aligned DEM. The *Height difference* view shows lower places in blue and higher places in red.

Confirm

Confirm writes the aligned map (files with the suffix `_coreg`, next to the originals) and **replaces the original map in the project**. The map keeps its name and id, so its colonies, comparison tables and exports stay the same. Its regions are moved with it; their surface areas and DEM metrics are cleared when the heights or the scale change (compute them again). If the aligned map reaches beyond the left or top edge of the reference, TagLab asks whether to **extend** the reference (it is padded with an exact shift) or **crop** the aligned map. Nothing is saved until you save the project.

8.3 Split screen

Compare ▸ Enable Split Screen (Alt + S) or the toolbar button shows two maps side by side.

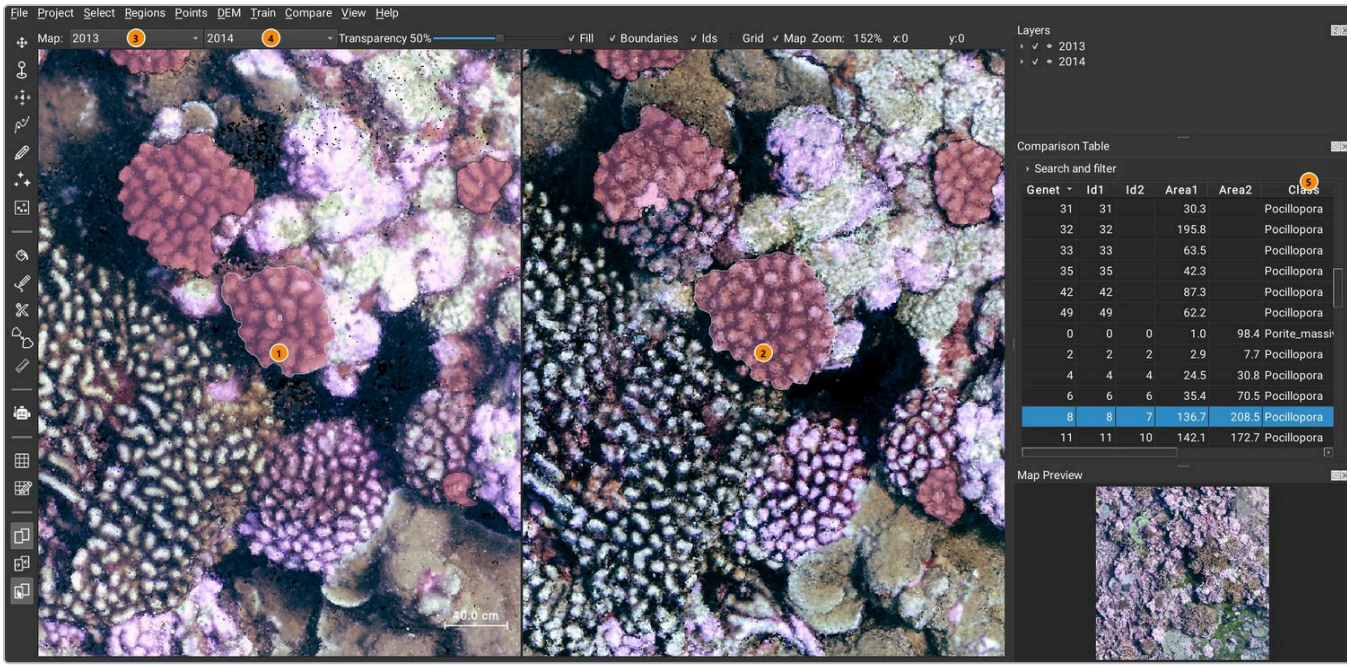


Figure 51. Split screen on the 2013 and 2014 maps.

1. The **older** map (the *source* of the comparison).
2. The **newer** map (the *target*).
3. The map shown on the left.
4. The map shown on the right: choose the pair here.
5. The **comparison table** of that pair, in place of the Data table.

Both maps move and zoom together. Clicking a row of the comparison table selects the colony on both maps. Every tool works on both maps; the map you last clicked is the *active* one (the menus and exports act on it).

8.4 Automatic matches

Compare ▸ Compute Automatic Matches (toolbar) matches the colonies of the two maps of the split screen. Two colonies are matched when they have the **same class** (not Empty) and they overlap by at least **60% of the smaller one**.

- If the pair already has a table, TagLab asks whether to clear it and replace all the matches.
- When the maps are not on the same pixel grid, TagLab says how it placed them (through the georeferencing, or an offset it found from the images) and asks: **Yes** uses that placement, **No** compares the maps as they are, **Cancel** stops. After the Alignment Tool the maps are on the same grid and the question doesn't appear.

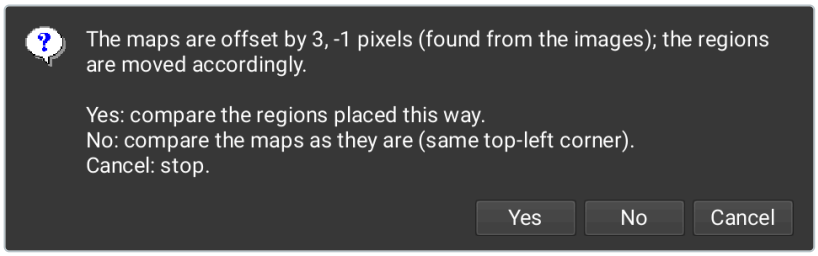


Figure 52. The placement question on the 2013 / 2014 sample, which has a small offset of its own.

Colonies of the old map without a match become **dead**, colonies of the new map without a match become **born**. A class change between the years shows up as dead + born: check those pairs.

8.5 Carrying regions over to a new survey

Compare ▸ Carry Regions Over... annotates the new map from the old one. Each colony of the old map is placed on the new map, at the same position or, if it was moved or turned (nursery bricks), where it is found nearby. Its outline is fitted to the new image, and TagLab checks whether the colony is still there (structure, colour, brightness, texture and, with DEMs, height).

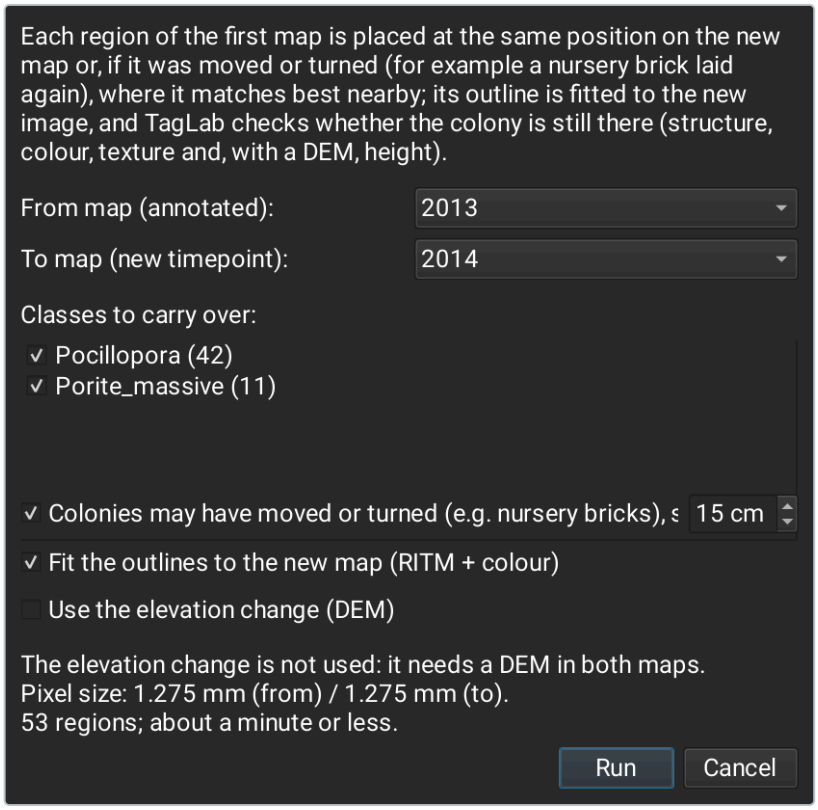


Figure 53. The Carry Regions Over window.

Option	Use
From map / To map	The annotated map and the new one (they must be aligned)
Classes to carry over	Untick classes you don't follow (e.g. Sand)
Colonies may have moved or turned	On for nursery bricks and anything that can be moved; search distance (default 15 cm)
Fit the outlines to the new map	On: the outline can shrink or grow (RITM + colour)
Use the elevation change (DEM)	Only when both maps have a DEM: a colony that disappeared also lowers the surface

Run (about 1 s per colony on an annotation computer). The new map gets every colony, with the same class, name and attributes, matched to its old one in the comparison table, and three new attributes:

Attribute	Values
`carryover`	`ok`, `check`, `possibly gone`, or `kept` after review
`carry_conf`	0 to 1: how sure TagLab is that the colony is still there
`carry_from`	The id of the colony on the old map

Nothing is deleted automatically. The doubtful colonies (`check` and `possibly gone`) open in the **review** window, most likely gone first, and each one is shown on both maps:

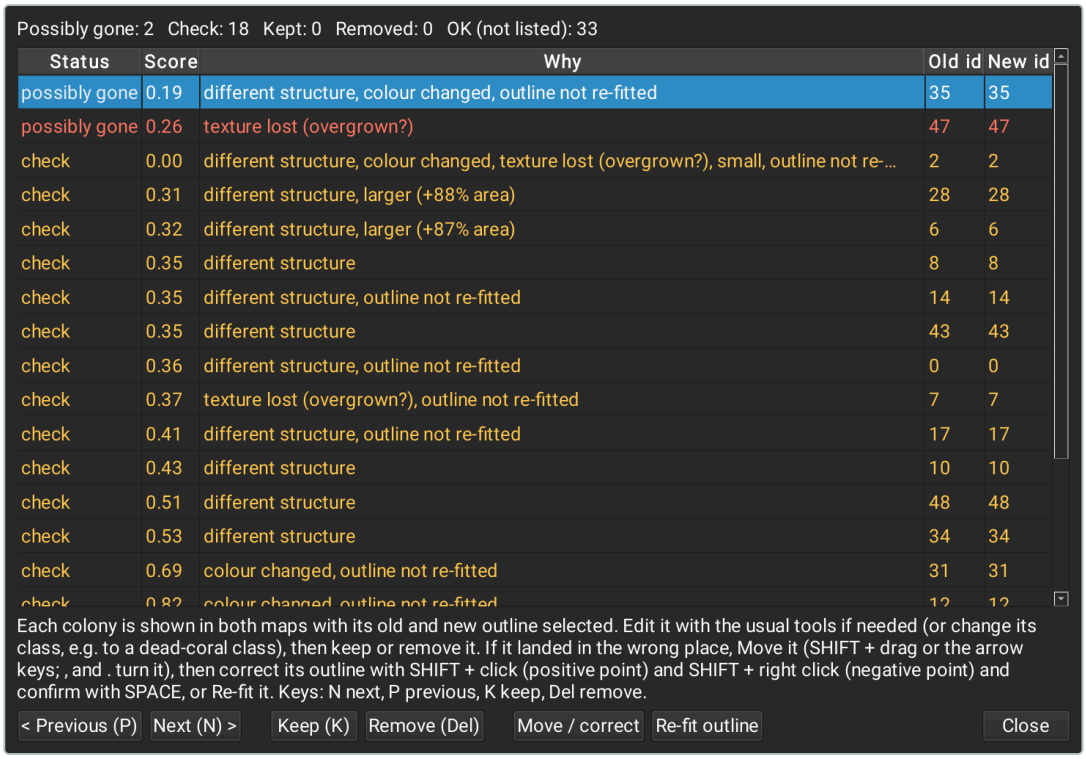


Figure 54. The review window after carrying the 53 colonies of 2013 to 2014: 33 ok (not listed), 18 to check and 2 possibly gone. The Why column gives the reasons.

Key / button	Action
N / P	Next / previous colony
K	Keep it (status `kept`)
Del	Remove it from the new map: it becomes *dead* in the comparison table
Move / correct	Selects the colony with the Move Region tool: drag it, turn it (, , .), correct its outline with points (section 4.10)
Re-fit outline	Fits the outline to the image again (after moving it)

You can also redraw the colony with any tool or change its class (for example to a dead-coral class). Moving or re-fitting doesn't change the status: press K to keep it. **Compare ▸ Review Carried-over Colonies** reopens the list. One Ctrl+Z removes everything the carry-over added.

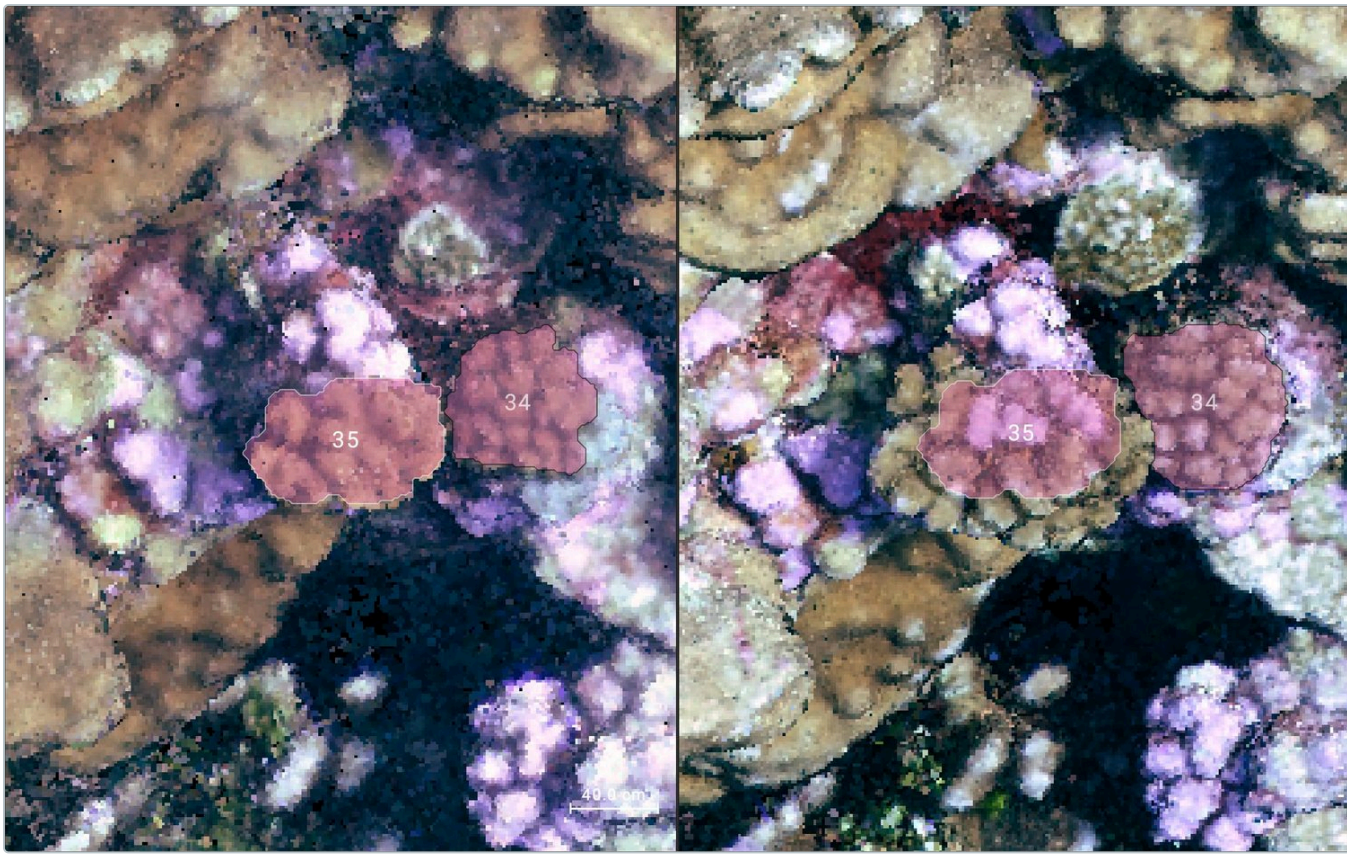
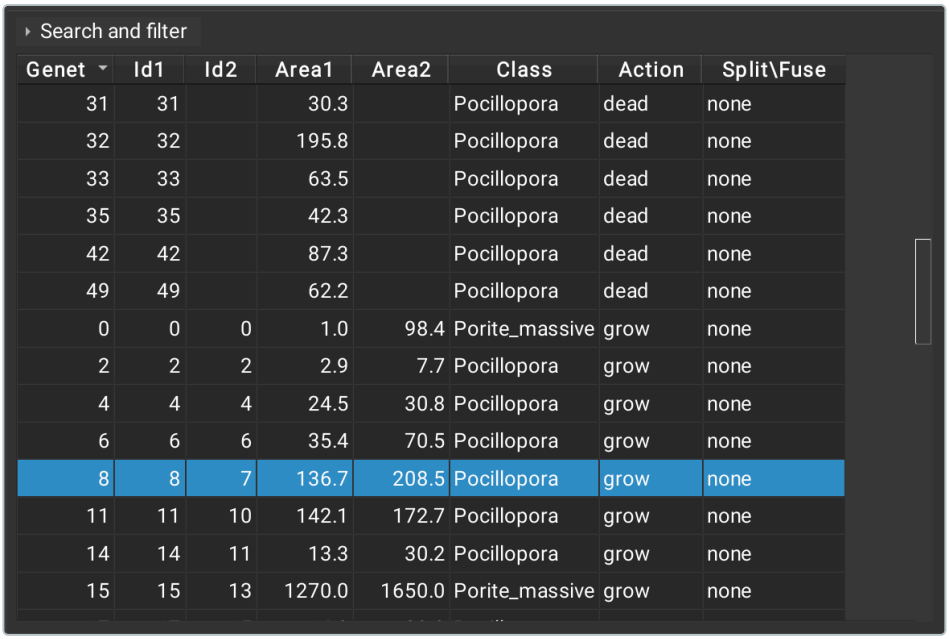


Figure 55. A colony under review: the old outline on 2013 (left) and the carried one on 2014 (right).

Check the ok colonies too

`ok` means that TagLab found the colony in place, not that the outline is perfect. Look through the new map with the fill off (Q) and correct outlines, then outline the **recruits** (new colonies), which carry-over cannot find.

8.6 The comparison table



Genet	Id1	Id2	Area1	Area2	Class	Action	Split\Fuse
31	31		30.3		Pocillopora	dead	none
32	32		195.8		Pocillopora	dead	none
33	33		63.5		Pocillopora	dead	none
35	35		42.3		Pocillopora	dead	none
42	42		87.3		Pocillopora	dead	none
49	49		62.2		Pocillopora	dead	none
0	0	0	1.0	98.4	Porite_massive	grow	none
2	2	2	2.9	7.7	Pocillopora	grow	none
4	4	4	24.5	30.8	Pocillopora	grow	none
6	6	6	35.4	70.5	Pocillopora	grow	none
8	8	7	136.7	208.5	Pocillopora	grow	none
11	11	10	142.1	172.7	Pocillopora	grow	none
14	14	11	13.3	30.2	Pocillopora	grow	none
15	15	13	1270.0	1650.0	Porite_massive	grow	none

Figure 56. The comparison table of 2013 → 2014.

Column	Meaning
Genet	The colony's lineage number
Id1, Id2	The colony's id in the left and the right map (empty for dead / born)
Area1, Area2	Its area in each map (cm^2; or surface area, see *Compare*)
Class	Its class
Action	Its fate (below)
Split\Fuse	*split*: one colony became several; *fuse*: several became one; *none*

Fate	Meaning
grow / shrink	The area changed by more than 5%
same	The area changed by 5% or less
dead	No match in the newer map (or set by hand on a matched colony that died in place)
born	No match in the older map (a recruit, or a colony missed in the older survey)

- Fates are always computed in cm^2 from the current outlines. For a split or fusion the whole group gets one fate.
- **Dark red rows** are *to check*: TagLab could not keep the match by itself (after an edit, a repair or a doubtful carry-over). Look at them and fix the match.
- **Search and filter** (click to open): *Filter* shows only one fate or the rows *To check*; *Compare* switches the areas between **Area** and **Surface Area** (computed in chapter 7); *Search Id1 / Id2* finds a colony.
- **Double-click the Action** of a matched colony to set it to *dead* (a colony that died in place: the skeleton is still outlined) or back to its computed fate.
- Click a row to select the colonies on both maps.

Search and filter

Genet	Id1	Id2	Area1	Area2	Class	Action	Split\Fuse
7	7		15.8		Pocillopora	dead	none
12	12		20.3		Pocillopora	dead	none
13	13		343.7		Porite_massive	dead	none
16	16		5.4		Pocillopora	dead	none
20	20		25.0		Pocillopora	dead	none
23	23		119.2		Pocillopora	dead	none
24	24		52.6		Porite_massive	dead	none
28	28		35.8		Pocillopora	dead	none
31	31		30.3		Pocillopora	dead	none
32	32		195.8		Pocillopora	dead	none
33	33		63.5		Pocillopora	dead	none
35	35		42.3		Pocillopora	dead	none
42	42		87.3		Pocillopora	dead	none
49	49		62.2		Pocillopora	dead	none

Figure 57. The same table filtered on Dead.

Editing matches by hand

Compare ▸ Edit Manual Matches:

- **Create a match:** SHIFT + click the colony on the left map and its match on the right map (SHIFT + click more colonies for a split or a fusion; SHIFT + click again to unselect one), then Space.
- **Delete a match:** select its row in the comparison table and press Del: the colonies become dead and born.

Creating or deleting matches by hand **cannot be undone**: save before you start.

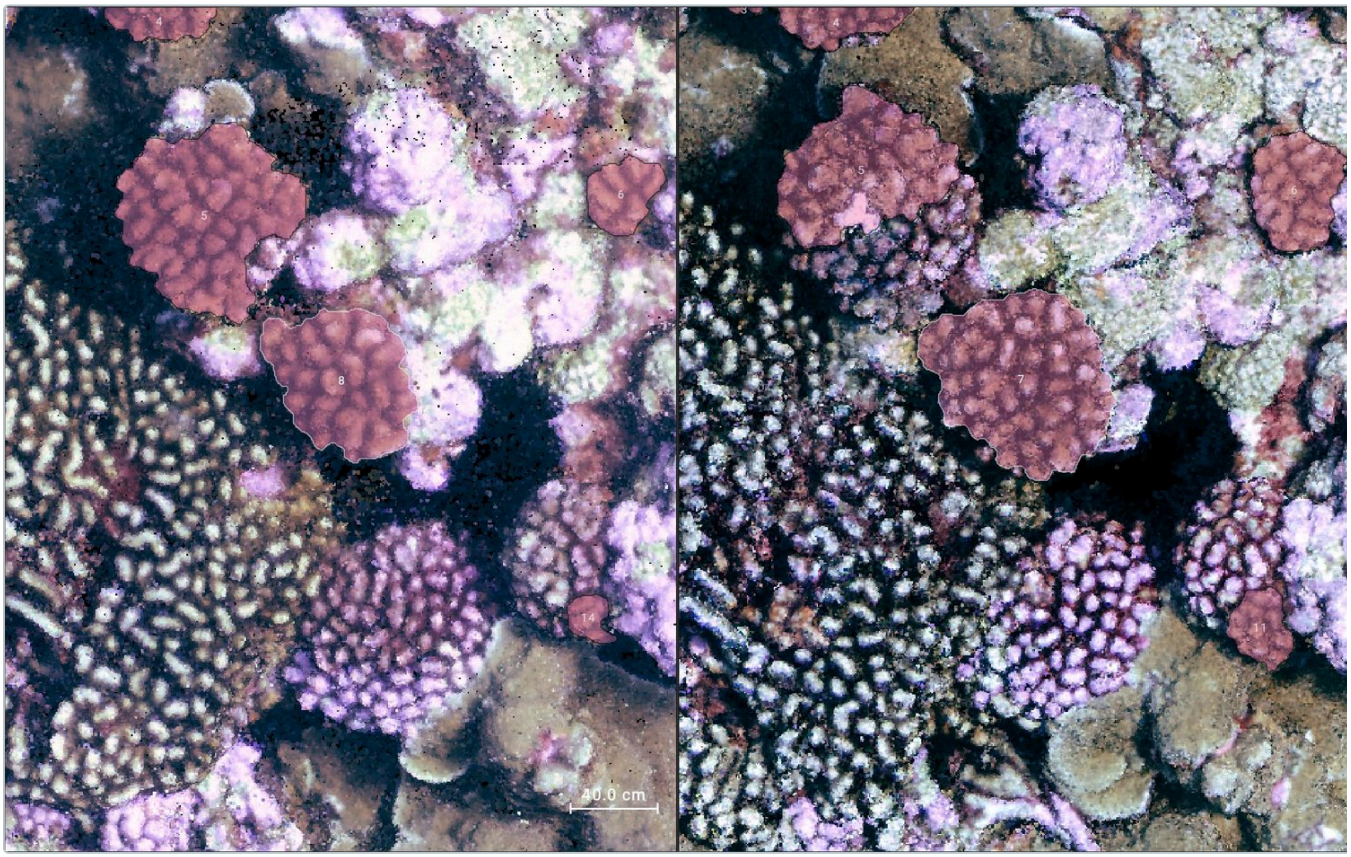


Figure 58. The Edit matches tool in split screen.

8.7 Exporting the year-to-year data

Compare ▸ Export Matches writes one CSV per pair of maps, ready for R or a spreadsheet:

Columns	Meaning
`colony`	Stable name of the colony: the map and id of its first region, e.g. `2013:45`
`genet`	TagLab's lineage number (can change after edits)
`map1, date1, id1, class1, area1_cm2, surf_area1_cm2`	The colony in the first map (surface area empty if not computed)
`map2, date2, id2, class2, area2_cm2, surf_area2_cm2`	The same in the second map
`fate`	grow / shrink / same / dead / born
`split_fuse`	none / split / fuse
`change_cm2, change_pct`	Area change; for splits and fusions, the change of the whole group
`years`	Time between the two dates, for annual rates
`to_check`	TRUE for the dark-red rows

Compare ▸ Export Genet Data as CSV writes one row per colony per map: `colony, genet, map, date, id, class, area_cm2, surf_area_cm2` (long format, for growth curves over many years).
Compare ▸ Export Genet Data As Shapes draws the outlines of each genet through the years in an SVG file.

Compare ▸ Clear Comparison Table deletes the table of the current pair (the regions stay).

8.8 SOP: annotating a new year

1. Save a copy of the project (**File ▸ Save As**, with the date in the name).
2. **Project ▸ Add New Map...**: the new orthomosaic and DEM, its date and pixel size.
3. **Project ▸ Alignment Tool**: reference = last year's map, image to align = the new map. Auto Markers, check the table, Preview, Check alignment, Confirm. Save.
4. Split screen: last year left, the new map right.
5. **Compare ▸ Carry Regions Over...**, then review every `check` and `possibly gone` colony (K / Del / Move / Re-fit).
6. Check the `ok` colonies and outline the recruits.
7. In the comparison table, look at *To check*, *Dead* and *Born*; fix the wrong matches.
8. Compute surface areas / DEM metrics on the new map if needed (chapter 7).
9. Save; export the data table of the new map and **Compare ▸ Export Matches**.

9. AI: training a network and classifying maps

TagLab can learn the lab's classes from maps that are already annotated, then classify new maps automatically. The network is a **DeepLab V3+** semantic segmentation model: it gives every pixel a class, and TagLab turns the classified pixels into regions. The result is a first draft: it always needs checking and correcting with the tools of chapter 4.

Step	Where	Computer
1. Export a training dataset from annotated maps	Train ▸ Export New Training Dataset	Any
2. Train a network on the dataset	Train ▸ Train Your Network	GPU computer
3. Classify a map, check and correct the regions	Toolbar (0), or `batch_processing.py`	GPU computer (a CPU works for small areas)

> **Sample data**
> The screenshots use the official VOS sample and a 2-epoch training on a few tiles, run only to show the windows. Its scores are meaningless (a real training runs for many epochs on the GPU computer).

9.1 Exporting a training dataset

The dataset is a folder of image **tiles** cut from the map, each with a **label image** in which every annotated pixel has the colour of its class and everything else is black (*Background*).

1. Open the annotated map. Annotate the area completely: every colony of the classes to learn must be outlined, or the network learns it as background.
2. Optionally set the working area to the annotated part.
3. Train ▸ Export New Training Dataset and fill in the window (below).
4. **Merge classes...** if needed (9.2), then **Export**. A message gives the folder and the number of classes, and the dataset folder gets a `classes.json` that lists them.
5. Save the project (the merges and any new labels are saved with it).

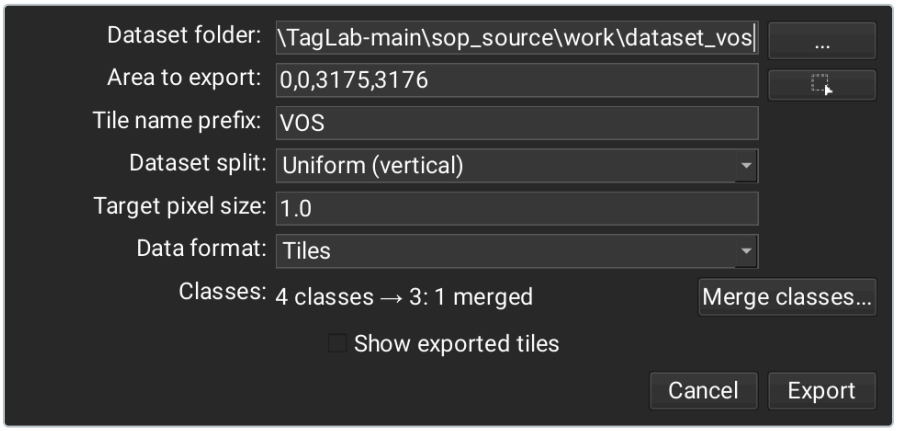


Figure 59. Export New Training Dataset.

Field	Meaning
Dataset folder	A new, empty folder for the dataset
Area to export	The part of the map (the working area by default); the button draws it on the map

Field	Meaning
Tile name prefix	Start of the tile names (the map name)
Dataset split	How the area is divided into training (70%), validation (15%) and test (15%) parts, below
Target pixel size	Pixel size (mm) of the tiles; the map is resampled to it. Keep the same value for all the datasets of a network
Data format	*Tiles* for Train Your Network; *COCO* or *YOLO-v5* for other programs
Classes	How many classes are exported, after merges

Dataset split:

- *Uniform (vertical)*: training is the top 70% of the area, then a band for validation and one for test.
- *Uniform (horizontal)*: the same from left to right.
- *Biologically-inspired*: TagLab searches for validation and test areas whose class composition is closest to the whole area, so rare classes are represented in all three parts. Best for maps where the classes are not spread evenly.

> **Black classes**
> Black is the Background colour of the label images. A class drawn in black (FDEP's *Sand*) would be confused with it: TagLab warns about it. Give such a class another colour (Labels Dictionary Editor) or merge it into another class before exporting.

9.2 Merging classes on export

Merge classes... gives each class an *Export as* name. Classes with the same name become one class in the tiles, for example *Vase (Boat)* and *Vase (Scuba)* → *Vase*: the network cannot tell how a sponge was planted, but the project keeps both classes for the CSVs and the year-to-year tracking.

- **Drop the (...) part of the names** does this for all the classes at once.
- *Background* leaves a class out (its pixels become black).
- A new name gets its own colour (the average of its classes, or click the colour to choose one); it must not be black or used by another label.
- The merges are saved with the project and reused at the next export. The new names are added to the project's labels.

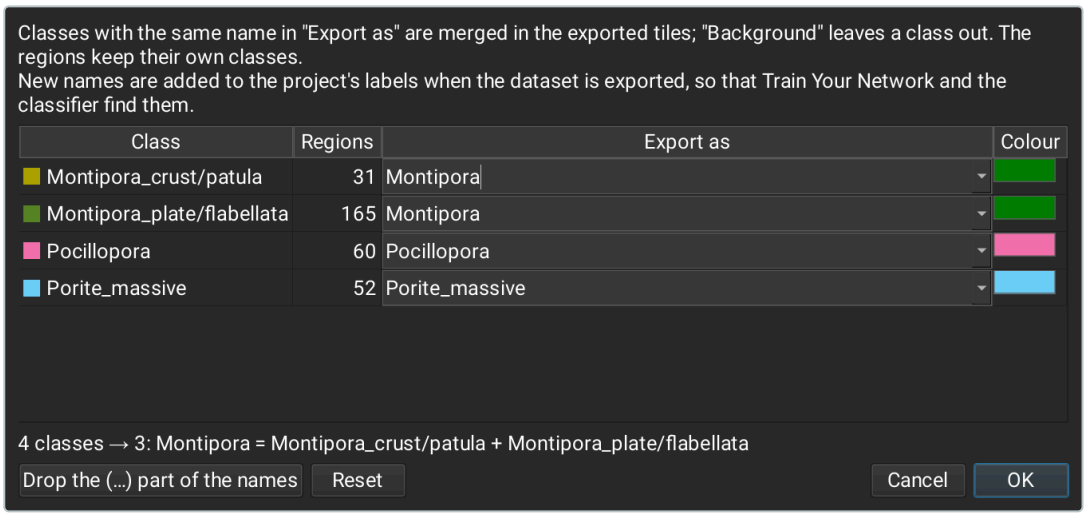


Figure 60. Merge Classes: the two Montipora classes of the sample exported as one class, "Montipora".

9.3 Training a network (GPU computer)

Train ▸ Train Your Network:

1. **Network name:** short and clear, for example `nursery_sponges_2026`. It becomes `models\<name>.net`.
2. **Dataset folder** (`...`): the exported folder. The classes and their share of the pixels are listed (from the dataset's `classes.json`). Untick a class to train without it (it is added to the background).
3. Set the parameters (below) and press **Train**. A progress bar shows the epochs; training takes from minutes to hours.
4. The **results** window opens at the end (9.4). **Confirm** adds the network to the list of classifiers; **Cancel** leaves it out (its file stays in `models\`).

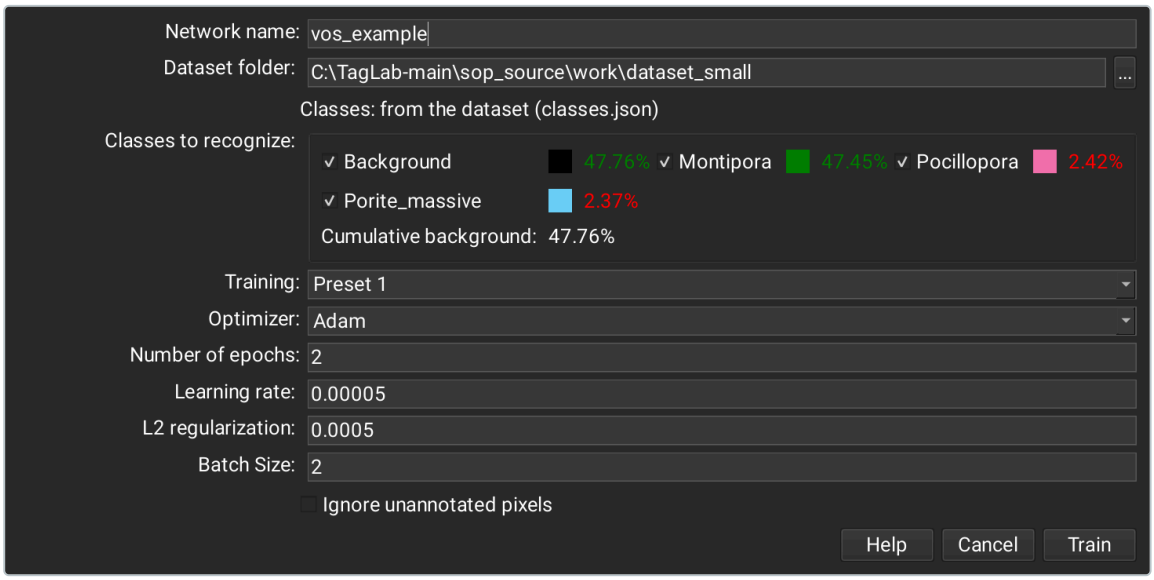


Figure 61. Train Your Network, with the classes read from the dataset's classes.json.

Parameter	Meaning	Suggested
Training	*Preset 1*: small adjustments to all the weights. *Preset 2*: the last layer first, then the decoder, then everything (epochs per stage); helps against overfitting on small datasets	Preset 1
Optimizer	Adam, or QHAdam (a variant)	Adam
Number of epochs	How many times the network sees every training tile	50–100
Learning rate	Size of each adjustment (Preset 1 only)	0.00005
L2 regularization	Penalty on large weights, against overfitting	0.0005
Batch size	Tiles processed together; limited by the GPU memory	8 (GPU), 4 (CPU)
Ignore unannotated pixels	Leaves black pixels out of the training instead of learning them as Background. Only for datasets where everything (sand, rubble, hard substrate…) is annotated, so the black pixels are just colonies that were missed	Off

9.4 Reading the training results

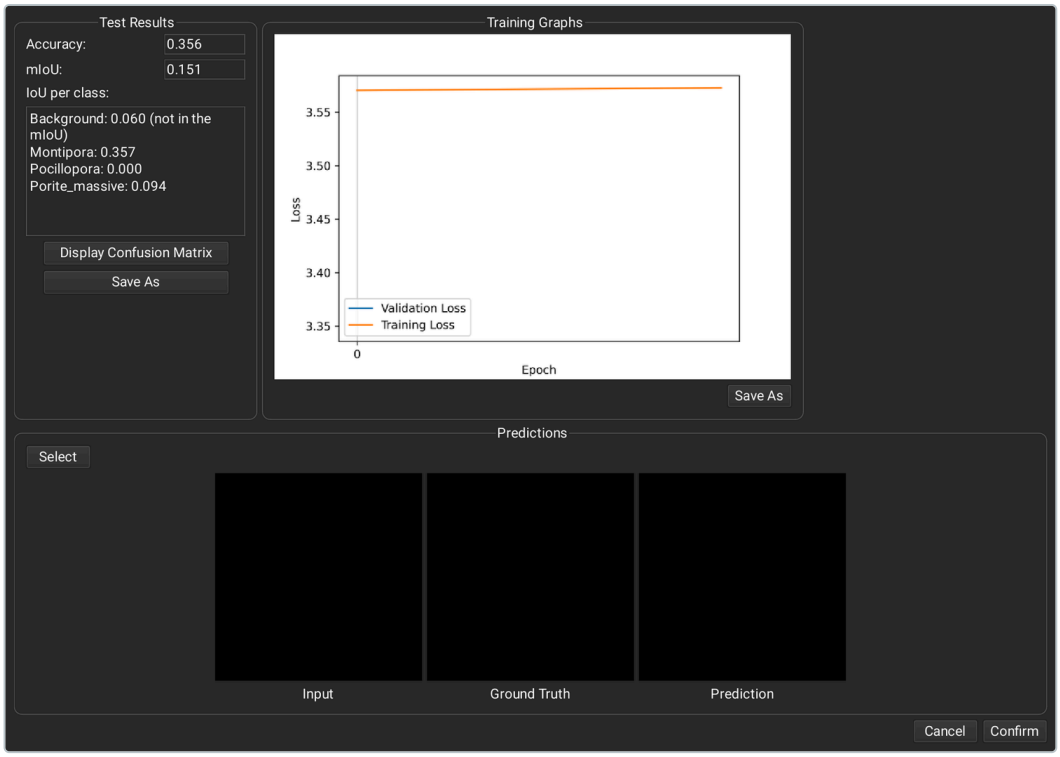


Figure 62. The results window of the 2-epoch example (its scores are meaningless).

- **Test Results** are measured on the *test* part of the dataset, which the training never used.
- **Accuracy:** the fraction of pixels classified correctly. It is misleading when one class (often the background) covers most pixels.
- **IoU per class** (intersection over union): the overlap between the predicted and the true pixels of a class, 0 to 1. **mIoU** is their mean over the classes (Background excluded): the main score. Training keeps the network of the epoch with the best mIoU on the validation part.
- **Display Confusion Matrix** shows which classes are mistaken for which.
- **Training Graphs:** the training loss (and the validation loss, computed every 2 epochs and at the end) should go down. A validation loss that rises while the training loss keeps falling means *overfitting*: the network memorises the training tiles.
- **Predictions:** *Select* a test tile to see it with its true labels and the network's prediction.
- As a rule of thumb, a class IoU above 0.7 gives good drafts, 0.5–0.7 usable drafts that need correcting, below 0.5 a class that needs more or better examples.

Discarding a test network: press **Cancel** in the results window, then delete `models\<name>.net` (about 230 MB). If it was confirmed, also delete its entry (from `{"Classifier Name": "<name>"` to the closing `}`) in `config.json`, with TagLab closed. Make a copy of `config.json` first.

9.5 Classifying a map

Toolbar (0) opens the classifier window.

1. **Classifier:** choose the network. Its file, classes and *training scale* (pixel size) are shown; the map is resampled to that pixel size.
2. **Check it on a small area first:** click the area button, drag a square on the map, then **Preview**. The right pane shows the classes found.

3. Adjust **Prediction threshold** (a pixel gets a class only when the network is at least that sure; higher = fewer, surer pixels) and **Transparency**. *Auto color* and *Auto contrast* correct the colours of a map that is much darker or bluer than the training maps.
4. **Apply** classifies the **working area** (or the whole map without one) in the background, with a progress window and Cancel. If some classes are not labels of the project, TagLab asks to add them.
5. The regions are added to the map as one undo step. Check and correct them, and outline what was missed.

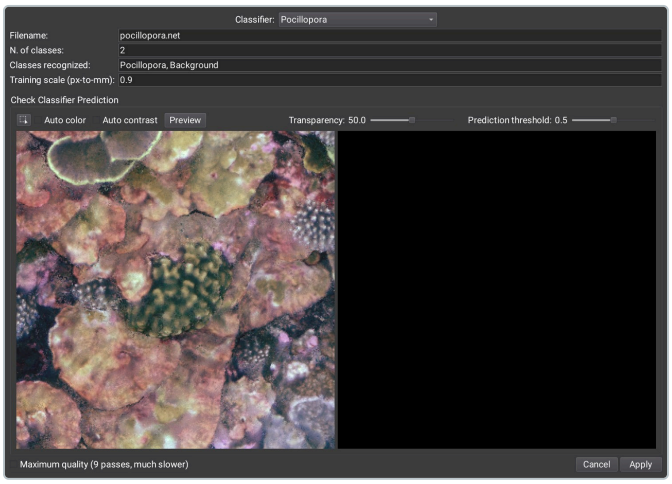


Figure 63. The classifier window with a preview area chosen.

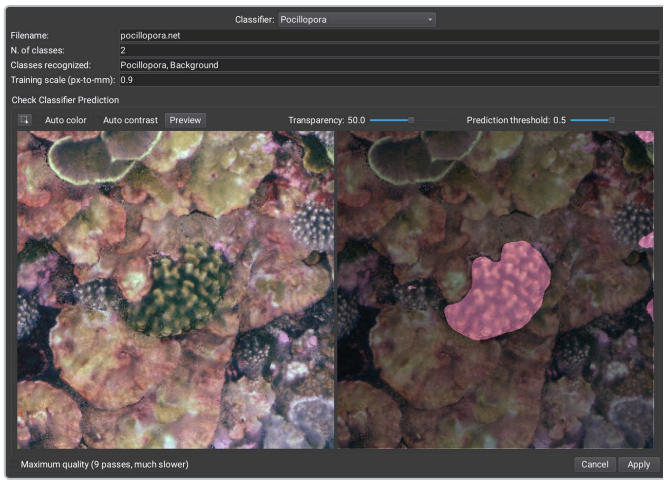


Figure 64. After Preview: the Pocillopora colonies found (pink).

- **Speed:** the normal (fast) mode is about 8 times quicker than the old method, with the same result on 99.5% of the pixels. On an annotation computer a 1500 × 1500 px area takes about 2 minutes; whole maps belong on the GPU computer. **Maximum quality** uses the old 9-pass method (much slower).
- **Classifiers:** the networks listed are those in `config.json`. A network trained with this version carries its class colours, so it can be applied to any project.

Classifying many maps: batch_processing.py

On the GPU computer, in a command window opened in the TagLab folder (use `venv\Scripts\python.exe` instead of `python` when TagLab has its own environment):

```
python batch_processing.py --projects_folder D:\Plots --classifier_name "Pocillopora" --output_folder D:\Classified
```

It classifies every map of every project (`.json`) in the projects folder, inside each project's working area, and saves the classified projects in the output folder (the originals are not changed). `--classifier_name` is the name shown in the classifier window. Options: `--autocolor`, `--autolevels`, `--quality` (maximum quality), `--batch-size N`, `--output-label-maps 0|1|2` (no label images / the working area / the whole map).

9.6 Good practice for training data

- **More examples beat more epochs.** Each class should have dozens of colonies over several maps, light conditions and years.
- **Annotate completely** inside the exported area: a colony left out is taught as background.
- **Merge classes the network can't see apart** in the image (9.2); keep them apart in the project.
- **Keep a test set** that the network never saw (the test split, or a separate map), and compare networks on it.
- **Keep the pixel size** the same for all datasets of one network.
- Write down the dataset, parameters and scores of each network you keep (lab log).

10. CoralNet

CoralNet classifies **points** on reef photos with a network trained on point annotations. TagLab can turn the lab's region annotations into CoralNet training data, and can bring CoralNet's predictions back as points.

10.1 Training a CoralNet source with your regions

1. **Points:** Points ▸ Sample Points On This Map (optional, section 5.2), then Points ▸ Label Points from Regions... (section 5.3). Every class gets at least the number of points you choose, away from the edges. Save the project.
2. **Export:** Points ▸ Export Tiled Data for CoralNet (below).
3. **Upload** to a CoralNet source whose label set has your codes: on the CoralNet website (upload the tiles, then the annotation CSV), or with the CoralNet Toolbox (10.2). **Try it on a test source first.**

Export Tiled Data for CoralNet cuts the map (or the working area) into tiles and writes:

- `tiles\`: the tiles that contain points (JPEG, quality 95);
- `coralnet_annotations.csv`: one row per point with a class, columns *Name, Row, Column, Label*, as CoralNet's annotation upload expects (Empty points are left out, and each point is in exactly one tile);
- `label_codes.csv`: each class, its CoralNet code and its number of points.

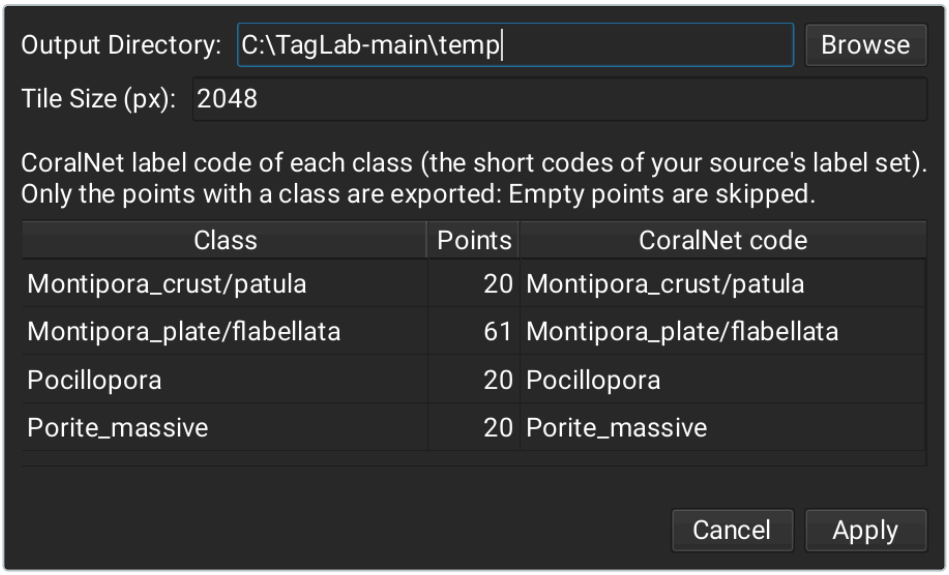


Figure 65. Export Tiled Data for CoralNet.

- **Output Directory:** where the export folder is created.
- **Tile Size:** the side of the tiles in pixels (224–8000, default 2048). CoralNet accepts images up to 8000 px.
- **CoralNet code:** the short code of each class in your CoralNet source's label set (for example `PAST` for *Porites astreoides*, if that is the code in your source). The codes are saved with the project and filled in the next time.

10.2 The CoralNet Toolbox

Points ▸ Open CoralNet-Toolbox... talks to CoralNet directly. Type your CoralNet **Username** and **Password** yourself and press **Authenticate**: the source list shows the sources you can use.

- **Upload Training Data...** (after Authenticate): choose a folder made by Export Tiled Data for CoralNet. It checks the folder, lists the codes, and asks before uploading the tiles and then the annotations to **Source ID 1**. Point annotations already on those images in the source are replaced.

- **Apply** (predictions): tiles the map, uploads the tiles to Source ID 1, asks the classifier of **Source ID 2** to classify the points, maps the CoralNet labels to TagLab classes (*Label Mapping*, which can be saved and loaded with Export / Import Mapping) and imports the result as points with machine suggestions (Info panel, *AI Suggestions*). Set both Source IDs with **Set Source IDs** first.

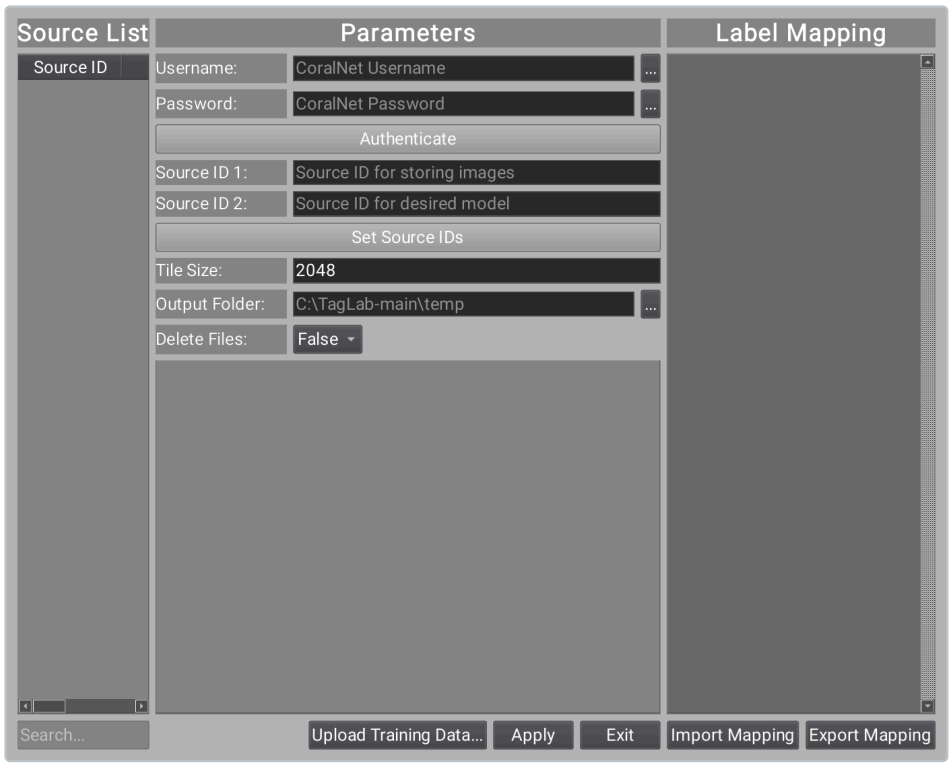


Figure 66. The CoralNet Toolbox before signing in.

> **Your CoralNet account**
>
> Only type your CoralNet password into the Toolbox yourself; never write it in a file or a message. Uploads go to the CoralNet source you choose and can replace its annotations: check the Source ID twice.

11. Settings and shortcuts

11.1 Settings

File ▸ Settings.. holds the computer's settings: the research field (*Marine Ecology*), **Autosave** (section 3.7), the default dictionary for new projects, and on the *Drawing* page the colours and widths of outlines, selections and the working area. They are stored on each computer, not in the project.

11.2 Keyboard and mouse shortcuts

File ▸ Keyboard and Mouse Shortcuts... changes any key of TagLab, and can put actions on the **middle button**, the two **thumb buttons** (back / forward) or the **right button** of the mouse. For example, put *Keep* of the carry-over review on the middle button to review colonies with one hand.

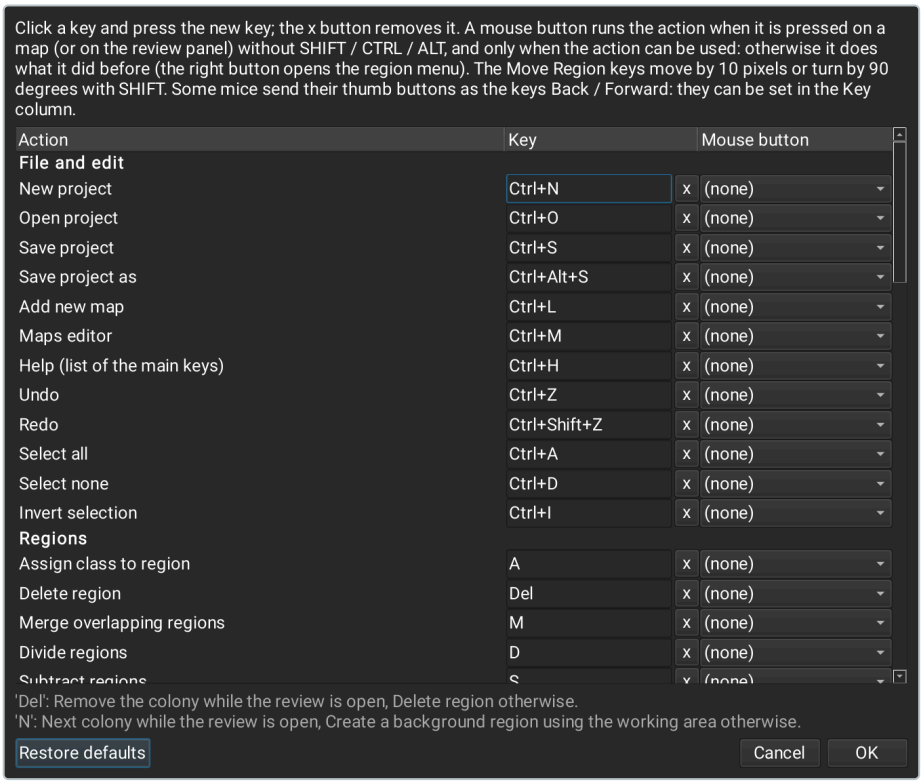


Figure 67. The shortcut editor: one row per action, grouped; a key cell and a mouse button for each.

- Click a **Key** cell and press the new key; × removes the key. Choose a **Mouse** button in the next column.
- **Conflicts** (the same key twice in a group) are shown in red and OK stays disabled until they are fixed.
- Space (confirm) and Esc (cancel) keep their keys; Tab, Home, End, [and] are reserved.
- A plain press of a mouse button (without SHIFT, Ctrl or Alt) on the map runs its action when the action can be used; otherwise the button works as before. SHIFT + click and SHIFT + right click (points) don't change.
- **Restore defaults** puts back the keys of this guide. The menus, the Help window and the tool instructions always show the keys in use.
- Shortcuts are stored on each computer. Appendix A lists the default keys.

12. Troubleshooting

12.1 When something goes wrong

1. **Read the message.** In the lab version an error does not close TagLab: a window says *An error occurred*, and the details go to `TagLab.log`.
2. **Save your work under a new name** (File ▸ Save As) before going on.
3. **Copy `TagLab.log`** (in the TagLab folder) somewhere safe: it is rewritten every time TagLab starts.
4. Write down what you did just before the error, and send both to the lab member in charge of TagLab.

12.2 Common problems

Problem or message	What to do
TagLab closes at start without a message	Double-click `opentaglab.bat` and read the black window. Run `update.bat`, then `install_windows.bat` again
A message at start says a newer version is available	Close TagLab and run `update.bat`
Locate the image when opening a project	The images were moved or renamed: point to the first one; the others in the same folder are found. Save the project
RITM or 4 clicks is slow, or runs out of memory	Zoom in: the tools work on the visible area
The outline of RITM keeps jumping	Esc, zoom closer, start with a point well inside the colony
A tool does nothing when you click	Hold SHIFT: tools act only on SHIFT + click. A plain drag pans
Georeferencing is not available	The map is not a GeoTIFF with coordinates: shapefile and GeoTIFF exports are not possible (section 6.4)
DEM not found	Add the DEM to the map in the Maps Editor (Depth Image)
Areas are clearly wrong	Check the pixel size of the map (Maps Editor, Ruler)
rows of the comparison tables ... were repaired when opening	Look at the dark-red rows (filter *To check*), fix them, save
The autosave cannot write its file	The folder is read-only or locked (OneDrive): TagLab keeps trying; check the folder, save by hand
Train Your Network lists only Background	The dataset has no `classes.json` and its colours don't match the project's labels: export the dataset again with this version
damaged or incomplete base network, or a model seems broken	`download_models.bat --check` lists every model as present, MISSING or INCOMPLETE; `download_models.bat` re-downloads the incomplete ones
The label X is missing when classifying	The classifier is older than this version: load the dictionary it was trained with

12.3 Keeping TagLab up to date

- `update.bat` updates TagLab from the lab's GitHub repository (`nirivas/TagLab-FIU`). It backs up every replaced file in `update_backups\` and never touches projects, data, models, `config.json` or trained networks.
- After an update that changes the required packages, run `install_windows.bat` again.
- `CHANGES.md` lists what the lab version changes; this guide is in `docs\`.

Appendix A. Default keys

The keys of a new installation. They can be changed on each computer (section 11.2); the menus and the Help window show the keys in use.

File and edit

New project	Ctrl + N
Open project	Ctrl + O
Save project	Ctrl + S
Save project as	Ctrl + Alt + S
Add new map	Ctrl + L
Maps editor	Ctrl + M
Help (list of the main keys)	Ctrl + H
Undo	Ctrl + Z
Redo	Ctrl + Shift + Z
Select all	Ctrl + A
Select none	Ctrl + D
Invert selection	Ctrl + I

Regions

Assign class to region	A
Delete region	Del
Merge overlapping regions	M
Divide regions	D
Subtract regions	S
Snap borders	B
Fill region	F
Refine border	R
Dilate border	+
Erode border	-
Create a background region using the working area	N

View

Split screen on / off	Alt + S
Switch RGB / DEM	C
Show / hide the fill of the regions	Q
Show / hide the borders	W
Show / hide the region ids	E
Show / hide the grid	G
Grid: change the state of the cell under the mouse	U

Tools

Pan / zoom	1
4-clicks segmentation	2
Positive / negative clicks segmentation (RITM)	3
Freehand segmentation	4
Assign class (tool)	5
Edit region border	6
Cut region	7
Create crack	8
Measure tool (ruler)	9
Fully automatic segmentation (classifier)	0
Move region	none
Place annotation point	none
Watershed segmentation	none
SAM: positive / negative clicks in an area	none
SAM: all instances in an area	none
Edit matches	none

Confirm and cancel

Confirm the operation of the tool	Space
Cancel the operation of the tool	Esc

Carry-over review

Next colony	N
Previous colony	P
Keep the colony	K
Remove the colony	Del

Move Region tool

Move left (SHIFT: 10 pixels)	Left
Move right (SHIFT: 10 pixels)	Right
Move up (SHIFT: 10 pixels)	Up
Move down (SHIFT: 10 pixels)	Down
Turn 5 degrees counter-clockwise (SHIFT: 90 degrees)	,
Turn 5 degrees clockwise (SHIFT: 90 degrees)	.
Turn 90 degrees counter-clockwise	<
Turn 90 degrees clockwise	>

Appendix B. Every menu command

File

- New Project Ctrl + N
- Open Project Ctrl + O
- Save Project Ctrl + S
- Save As Ctrl + Alt + S
- Import
  - Import Label Image
  - Import Shapefile
  - Add Another Project
- Export
  - Export Annotations As Data Table
  - Export Regions As Labeled Image (PNG)
  - Export Regions As Shapefile
  - Export Regions As DXF
  - Export Regions As SVG
  - Export Regions As A GeoTiff
  - Export Orthoimage As A GeoTiff
  - Export data for Reef Archive Lab platform
  - Export Histogram From Regions
  - Export New Training Dataset
- Settings..
- Keyboard and Mouse Shortcuts...

Project

- Add New Map... Ctrl + L
- Maps Editor... Ctrl + M
- Georeferencing Tool
- Set Working Area
- Alignment Tool
- Labels Dictionary Editor...
- Region Attributes...

Select

- Select All Ctrl + A
- Select None Ctrl + D
- Invert Selection Ctrl + I
- Select by current Class
- Select by Working Area
- Select by Properties

Regions

- Undo Ctrl + Z
- Redo Ctrl + Shift + Z
- Assign Class to Region A
- Delete Region Del
- Merge Overlapping Regions M
- Divide Regions D
- Subtract Regions S
- Snap Borders B
- Fill Region F
- Refine Border R
- Refine All Borders
- Dilate Border +
- Erode Border -
- Create a Background Region using the WA N
- Compute Geometric Info
- Course Analysis

Points

- Sample Points On This Map
- Label Points from Regions...
- Import Viscore Point Annotations
- Import CoralNet Point Annotations
- Export CoralNet Point Annotations
- Export Tiled Data for CoralNet
- Open CoralNet-Toolbox...

DEM

- Switch RGB/DEM
- Calculate Surface Area
- Calculate DEM Metrics
- Export Plot Metrics (CSV)
- Export Clipped Raster

Train

- Export New Training Dataset
- Dataset Manager
- Train Your Network

Compare

- Enable Split Screen Alt + S
- Compute Automatic Matches
- Edit Manual Matches
- Export Matches
- Export Genet Data As Shapes
- Export Genet Data as CSV
- Carry Regions Over...
- Review Carried-over Colonies
- Clear Comparison Table

View

- Labels
- Layers
- Info and Attributes
- Map Preview
- Data table

Help

- Help Ctrl + H
- Learn TagLab
- Report Issues
- About TagLab

Appendix C. Units and formulas

With s = the pixel size of the map in mm (Maps Editor):

Value	Formula
Area (cm^2)	pixels × s^2 / 100
Perimeter (cm)	pixel perimeter × s / 10
Length (Ruler, mm)	pixels × s
Surface area (cm^2)	Σ over the region's pixels of s^2 / cos(slope) / 100 (slopes over 87° count as flat)
Rugosity	surface area / planar area (1 = flat)
Grow / shrink / same	grow if area2 > 1.05 × area1, shrink if area2 < area1 / 1.05, otherwise same (areas in cm^2; for a split or fusion, the sums of the group)
`change_cm2`, `change_pct`	area2 − area1, and (area2 − area1) / area1 × 100
`years`	(date2 − date1) in days / 365.25

- The pixel size is **never** read from the image file: it is the value typed in the map settings (or set with the Ruler).
- Centroids and the *Area to export* / working area are in **pixels** of the map, measured from its top-left corner.
- DEM elevations are in the units of the DEM (normally metres); `relief_cm`, `height_cm`, `volume_cm3` convert them.

Appendix D. Columns of the data table export

File ▸ Export ▸ Export Annotations As Data Table writes one row per visible region and/or point (inside the working area when one is set):

Column	Content
`Image name`	File name of the map's orthomosaic
`TagLab Date`	Acquisition date of the map
`TagLab Type`	Region or Point
`TagLab Genet Id`	Lineage number across the years (0 when there is none)
`TagLab Id`	Id of the region or point (the number drawn with `E`)
`TagLab Class name`	Class
`TagLab Centroid x`, `TagLab Centroid y`	Centre of the region (pixels)
`TagLab Perimeter`	cm
`TagLab Area`	cm^2
`TagLab Surf. area`	cm^2 (0 when not computed)
`TagLab Note`	The region's note
one column per region attribute	Including the DEM metrics (`slope_deg`, `elev_mean`, …) and the carry-over values (`carryover`, `carry_conf`, `carry_from`)

The shapefile export has the same values as attribute fields (names cut to 10 characters).

Glossary

Maps and projects

Project
A TagLab `.json` file. It holds the list of maps (with the paths of their image files), the labels, every region and point, the comparison tables and the settings of the project. The images are not inside it: keep the images where the project expects them, or TagLab asks for them when it opens the project.

Map
One survey of one plot in a project: an orthomosaic, optionally with its DEM, a name, a pixel size and an acquisition date. Called *image* in some TagLab messages.

Plot
The area of reef surveyed and compared over time (for example a Culebra year-to-year plot or a PEV plot).

Orthomosaic (ortho)
A single top-down image of the plot, stitched from many photos by photogrammetry (Metashape, Viscore…), with the same scale everywhere.

DEM (digital elevation model)
A raster of heights (one value per pixel, in metres) made with the orthomosaic. TagLab uses it for surface area, DEM metrics, plot rugosity and the height checks of the Alignment Tool and Carry Regions Over.

Channel
One image of a map: **RGB** (the orthomosaic) or **DEM**. C switches the view between them.

Pixel size
The size of one map pixel on the reef, in millimetres (mm/px). Every area, perimeter and length in cm depends on it. TagLab never reads it from the GeoTIFF: type it in Map Settings or measure it with the Ruler.

Acquisition date
The survey date of a map (YYYY-MM-DD). The year-to-year exports use it to compute the time between two maps.

Georeference
The position of the map in real-world coordinates, stored in a GeoTIFF. Needed by the shapefile and GeoTIFF exports; when both maps of a pair have one, it is also used to line them up.

GeoTIFF
A TIFF image with a georeference. Orthomosaics and DEMs exported by photogrammetry software are usually GeoTIFFs.

Working area
A rectangle of the map set with Project ▸ Set Working Area. Exports, DEM plot metrics, Label Points from Regions and the training dataset can be limited to it. Not the same as the *work area* of RITM or SAM.

Work area (RITM, SAM)
The part of the map an AI tool looks at: for RITM, the view at the first click; for SAM, the square chosen in step 1. Keep it small: larger areas are slower and less precise.

Grid
A grid of cells drawn on the map (toolbar button *Create grid*), to annotate a large map cell by cell. Each cell can be marked empty, incomplete or complete, with a note.

Sampling area
An area where Points ▸ Sample Points On This Map places points (a single rectangle, the working area, or transects).

Layer
An extra drawing over a map (for example an imported shapefile, the grid or sampling areas), listed in the Layers panel.

Label, class
The category a region or point belongs to (a species, Sand, Brick…), with a fill colour. The project's list of labels is shown in the Labels panel.

Dictionary
A `.json` file with a list of labels and their colours (for example `FDEP.json`, `spongenursery.json`). It is copied into the project when you load it; later changes to the file are not synced.

FDEP list
The Florida Department of Environmental Protection species list used by the lab (`FDEP.json`).

Region attributes
Extra fields that every region can have (numbers, text or keyword lists), defined in Project ▸ Region Attributes.... DEM metrics and the carry-over status are stored as region attributes.

Autosave file
`<project>_autosave.json`, written next to the project every few minutes when autosave is on. It never replaces the project file.

`TagLab.log`
The log file in the TagLab folder. Errors, timings and the results of long operations are written there.

`config.json`
TagLab's settings file in the TagLab folder. It lists the available classifiers (networks).

Regions and annotation

Region
One outlined object on a map (a colony, a sponge, a patch of sand...): an outline, optional holes, a class, an id and measurements. Called *blob* in the code.

Outline (contour)
The border of a region. Holes inside a region are *inner contours*.

Id
The number of a region in its map. Ids are never reused, so a deleted id stays unused.

Empty
The class of a new region before a class is assigned. Empty regions are skipped by auto-match and by most exports.

Point (annotation point)
A single annotated position with a class, as in point-count surveys and CoralNet.

Selection
The regions (and points) currently highlighted. Operations such as Merge, Delete or Calculate DEM Metrics work on the selection.

Active map
In split screen, the map you clicked last. Menu commands work on it.

Split screen
Two maps side by side (usually two years of the same plot), each with its own view.

Tool instructions
The panel at the bottom left of the map that lists the mouse and keys of the current tool.

Positive / negative point
A click that tells an AI tool "this is part of the object" (positive, SHIFT + left click) or "this is not" (negative, SHIFT + right click).

RITM
The positive/negative clicks tool: an AI model that outlines an object from a few clicks. It also corrects an existing region. Works well on a CPU.

DEXTR (4 clicks)
An AI model that outlines an object from its four extreme points (top, bottom, left, right).

SAM (Segment Anything)
A large AI model that outlines every object in an area, or one object from clicks. Very slow without an NVIDIA GPU, and optional at install time.

Refine Border
Moves the border of the selected region onto the nearest strong edges of the image.

Snap Borders
Makes the borders of touching regions follow each other exactly (no gaps or overlaps).

Dilate / Erode
Grow or shrink the selected regions by one pixel all around.

Background region
New regions (class Empty) that cover every part of the working area without a region (**Regions ▸ Create a Background Region using the WA**). Give them a class such as Sand.

Background (training)
In a training dataset, the pixels that belong to no exported class (black in the label images).

Measurements

Area
The planar (top-down) area of a region in cm^2: pixels × (pixel size)^2 / 100.

Perimeter
The length of the outline in cm: pixels × pixel size / 10.

Surface area
The 3D area of the region measured on the DEM (always larger than the planar area). Computed with **DEM ▸ Calculate Surface Area**.

Slope
The steepness of the reef surface, in degrees from horizontal, computed from the DEM with Horn's method (as `gdaldem slope`).

Rugosity
Surface area divided by planar area: 1 for a flat surface, larger for a rougher one. The value depends on the DEM resolution, so compare plots with `rugosity_1cm`.

`rugosity_1cm`, `slope_1cm`
Plot rugosity and slope computed on the DEM averaged to 1 cm cells. Use them to compare surveys whose DEMs have different resolutions.

Relief
For a colony: the height range inside its outline (cm). For the plot: the height range after removing the plot's overall tilt (1st to 99th percentile).

Height above plane, volume
The height (cm) and volume (cm^3) of a colony above a plane fitted to the substrate just outside its outline.

Elevation
The DEM value (metres) inside a region: `elev_mean`, `elev_min`, `elev_max`.

DEM metrics
The values stored per region by **DEM ▸ Calculate DEM Metrics**: `slope_deg`, `elev_mean`, `elev_min`, `elev_max`, `relief_cm`, `height_cm`, `volume_cm3`, `rugosity`. Regions with less than half of their area covered by the DEM get no values.

Plot metrics
The rugosity, slope and relief of the whole plot (the working area or the whole map), stored with the map and exported by **DEM ▸ Export Plot Metrics (CSV)**.

Centroid, bounding box
The centre of a region, and the smallest rectangle that contains it (in map pixels).

Year-to-year

Timepoint
One survey of a plot (one map in the project).

Alignment (co-registration)
Transforming the new map so that it lies exactly on the reference map, pixel for pixel. Done with the Alignment Tool before comparing years.

Reference map, aligned map
In the Alignment Tool, the map that stays as it is (reference, left) and the map that is moved onto it (right).

Marker
A pair of points, one on each map, on the same feature. The alignment is fitted to the markers. A *hard* marker (red; SHIFT + click on a marker toggles it) counts three times as much as a *soft* one in the fit.

Auto Markers
Finds markers automatically by matching image features (and, with DEMs, the relief), whatever the rotation.

RMSE
Root mean square error: the typical distance (mm) between where the markers are and where the fitted alignment puts them. Small is good; a marker with a large error is shown in red.

Apply scale
Lets the alignment also change the size of the moved map (when the two surveys have slightly different scales). Off by default (rotation and shift only).

Check alignment, quality map
A measurement, in the Alignment Tool preview, of the misalignment left after the fit, on a grid of cells: green ≤ 1 mm, yellow ≤ 3 mm, red > 3 mm, grey not measurable.

Comparison table
The table of matches between two maps (one table per pair), shown in split screen in place of the data table.

Match
A link between a region of the first map and a region of the second map: the same colony in the two surveys.

Fate
What happened to a colony between the two maps, in the Action column: **grow** / **shrink** (area changed by more than 5%), **same**, **dead** (no match in the second map, or set by hand), **born** (no match in the first map). **n/s** (not set) appears only for a moment, before the areas are computed.

Split, fuse
One colony matched to several (split) or several matched to one (fuse). The whole group gets one fate.

Genet
TagLab's lineage number: all the regions linked by matches across the maps form one genet. It is recomputed when matches change.

Colony name
A stable name for a colony in the exports, `map:id` of its first region (for example `2013:45`).

To check
A match TagLab could not keep with certainty (after an edit, a repair or a doubtful carry-over), shown in dark red.

Placement
How Carry Regions Over and auto-match line up two maps: *same grid*, *georeference*, or an *offset* found from the images.

Carry-over
Copying the regions of an annotated map onto the next survey (Compare ▸ Carry Regions Over...), refitting their outlines and checking whether each colony is still there.

Carry-over status
The `carryover` attribute of a carried region: **ok**, **check** (look at it), **possibly gone**, or **kept** (confirmed in the review). `carry_conf` is the confidence (0–1) and `carry_from` the id on the old map.

AI

Training dataset
Image tiles and their label images exported from annotated maps (Train ▸ Export New Training Dataset), split into training, validation and test sets.

Tile
A square piece of the map (513 × 513 pixels for TagLab's networks) used for training and classification.

Target pixel size
The pixel size the tiles are resampled to. A classifier works best on maps with the pixel size it was trained on.

Dataset split
How the map is divided into training, validation and test areas: *uniform* (vertical or horizontal strips) or *biologically-inspired* (areas chosen so that each set has a similar mix of classes).

Training / validation / test set
Training tiles teach the network; validation tiles choose the best epoch; test tiles give the final, independent score.

`classes.json`
The file in an exported dataset that lists each class, its colour and the classes it was merged from. Train Your Network reads the classes from it.

Class merge
Exporting several classes under one name (for example "Vase (Boat)" + "Vase (Scuba)" → "Vase") without changing the regions.

Network, classifier (`.net`)
A trained model saved in `models/`. Classifiers are listed in `config.json` and chosen in the Select Classifier window.

DeepLab V3+
The type of semantic segmentation network TagLab trains (with a ResNet-101 backbone).

Semantic segmentation
Giving a class to every pixel of the map. TagLab then turns the classified pixels into regions.

Epoch
One pass of the training over all the training tiles.

Batch size
How many tiles the network sees at once. Larger batches need more GPU memory.

Learning rate
How large the steps are when the network's weights are adjusted.

L2 regularization
A penalty on large weights that reduces overfitting.

Preset
Training schedule: *Preset 1* adjusts all the weights of the network; *Preset 2* trains the last layer first, then the decoder, then the whole network.

Loss
The training error the network minimises. It should go down over the epochs for both training and validation.

Accuracy
The fraction of pixels classified correctly. It is dominated by the common classes (often the background).

IoU, mIoU
Intersection over union: the overlap between predicted and true pixels of a class, divided by their union (1 = perfect). mIoU is the mean over the classes (Background excluded); TagLab keeps the epoch with the best mIoU.

Confusion matrix
A table of how many pixels of each true class were predicted as each class.

Overfitting
When the network learns the training tiles by heart and does worse on new ones: training loss keeps going down while validation loss goes up.

Augmentation
Random changes to the training tiles (flips, rotations, colour changes, small holes) so that the network generalises better.

Ignore unannotated pixels
A training option: pixels with no label are left out instead of being taught as background. Use it only for areas annotated completely.

Preview
Classifying a small area in the Select Classifier window, to check a network before classifying the whole map.

Fast / Maximum quality
The two classification modes: overlapping tiles with smooth blending (default), or the old 9-pass method (about 8 times slower, nearly the same result).

CPU, GPU, CUDA
The processor and the graphics card. Training and SAM need an NVIDIA GPU (CUDA) to be practical; annotation tools such as RITM run fine on the CPU.

`batch_processing.py`
A script that classifies all the projects of a folder with a classifier, without the window (for the GPU PC).

CoralNet, source, label set
CoralNet is the online point-annotation and machine-learning platform. A *source* is a CoralNet project; its *label set* lists the labels it uses, each with a short *label code*.

CoralNet Toolbox
The window (Points ▸ Open CoralNet-Toolbox...) that signs in to CoralNet and uploads tiles and annotations or gets predictions.

Software

`venv`
: TagLab's own Python environment, in the `venv` folder of the TagLab folder.

`opentaglab.bat`
: Starts TagLab (the desktop shortcut points to it).

`install_windows.bat`
: Installs the Python packages, PyTorch (GPU version when there is an NVIDIA card) and the AI models.

`update.bat`
: Updates TagLab from the lab's GitHub repository. Replaced files are saved first in `update_backups\`.

`download_models.bat`
: Downloads missing or incomplete AI models; `--check` only lists them.

Shortcut editor
: **File ▸ Keyboard and Mouse Shortcuts...**: changes any key, or puts an action on a mouse button.